

More Efficient Isogeny Proofs of Knowledge via Canonical Modular Polynomials

Thomas den Hollander^{*} Sören Kleine[†] Marzio Mula[‡] Daniel Slamanig[§]
Sebastian A. Spindler[¶]

Research Institute CODE, Universität der Bundeswehr München

Abstract

Proving knowledge of a secret isogeny has recently been proposed as a means to generate supersingular elliptic curves of unknown endomorphism ring, but is equally important for cryptographic protocol design as well as for real world deployments. Recently, Cong, Lai and Levin (ACNS'23) have investigated the use of general-purpose (non-interactive) zero-knowledge proof systems for proving the knowledge of an isogeny of degree 2^k between supersingular elliptic curves. In particular, their approach is to model this relation via a sequence of k successive steps of a walk in the supersingular isogeny graph and to show that the respective j -invariants are roots of the second modular polynomial. They then arithmetize this relation and show that this approach, when compared to state-of-the-art tailor-made proofs of knowledge by Basso et al. (EUROCRYPT'23), gives a 3-10 $\times$ improvement in proof and verification times, with comparable proof sizes.

In this paper we ask whether we can further improve the modular polynomial-based approach and generalize its application to primes $\ell > 2$, as used in some recent isogeny-based constructions. We will answer these questions affirmatively, by designing efficient arithmetizations for each $\ell \in \{2, 3, 5, 7, 13\}$ that achieve an improvement over Cong, Lai and Levin of up to 48%.

Our main technical tool and source of efficiency gains is to switch from classical modular polynomials to canonical modular polynomials. Adapting the well-known results on the former to the latter polynomials, however, is not straight-forward and requires some technical effort. We prove various interesting connections via novel use of resultant theory, and advance the understanding of canonical modular polynomials, which might be of independent interest.

1 Introduction

More than twenty years have passed since the seminal works by Couveignes [Cou06], Rostovstev, and Stolbunov [RS06] have introduced the idea of using maps between elliptic curves, called *isogenies*, for cryptographic purposes. Although their original attempts seemed too inefficient to compare with concurrent cryptosystems, later efforts in this direction [JD11, CLM⁺18] gave birth to a rich, and still lively, branch of cryptography.

A strong reason for researchers to push into this field is that the main problem on which it is based – namely, recovering a secret isogeny between two given elliptic curves – is considered hard even for quantum computers. Moreover, compared with other proposals for post-quantum cryptography, isogenies enjoy shorter parameters which though come at the price of slower performance. Since its proposal, isogeny-based cryptography has evolved into a very active and dynamic field, and many different cryptographic applications have been proposed so far.

In this work we are focusing on non-interactive zero-knowledge proofs of knowledge of secret isogenies, which are an important tool for cryptographic protocol design.

^{*}thomasdh@unibw.de

[†]soeren.kleine@unibw.de

[‡]marzio.mula@unibw.de

[§]daniel.slamanig@unibw.de

[¶]s.spindler@unibw.de

© IACR 2025. This article is the full version of the article published by Springer-Verlag available at https://doi.org/10.1007/978-3-032-01855-7_5.

An immediate application of such proofs is the design of signature schemes obtained via the Fiat-Shamir heuristic, e.g. GPS signatures [GPS17], CSI-FiSh [BKV19] or SQISign [DKL⁺20]. Moreover, they can be used to construct related primitives such as verifiable random functions (VRFs), as recently demonstrated using isogenies by Levin and Pedersen [LP24].

They also enable cryptographic tools that otherwise require a trusted setup. More precisely, such proofs have been studied for settings where one wants to avoid a trusted setup to generate supersingular curves of unknown endomorphism ring [BCC⁺23, CLL23]. Such curves are needed for several isogeny-based protocols ranging from hash functions [CLG09] to verifiable delay functions (VDFs) [BBBF18, DMPS19], delay encryption schemes [BD21], and public-key encryption [FMP23, Mor23]. In all these applications it is central to the security that the trapdoor is discarded after the trusted setup – a requirement that is hard to enforce in practice. Basso et al. [BCC⁺23] propose to implement a sequential multi-party ceremony to replace the trusted setup. Loosely speaking, they consider a walk in the isogeny graph

$$E_0 \rightarrow E_1 \rightarrow \cdots \rightarrow E_k$$

that starts from some (known) curve E_0 and then each party i takes the previous curve E_{i-1} , generates a random isogeny to a new curve E_i and provides a proof that they know the isogeny from E_{i-1} to E_i . Such a protocol can be used to replace a trusted setup as long as one of the parties in the chain can be assumed to be honest (i.e. discards its secret isogeny).¹

Finally, such proofs are a central tool to enforce honest behavior in multi-party protocols, where parties are forced to demonstrate that certain values such as (partial) public keys are well formed, e.g. in distributed key generation schemes [ABCP23a, ABCP23b]. These proofs-of-possession (PoPs) are, moreover, an important measure to prevent rogue key attacks in multi-party signature protocols [RY07]. In practice, this is required in public-key infrastructures (PKIs) when requesting the issuance of a certificate. In current PKIs based on X.509 certificates [CSF⁺08], these so-called certificate signing requests realize these PoPs via signatures.

While this only works for certifying signing keys, it becomes more relevant in a post-quantum setting – for example, when certifying KEM keys for KEMTLS [SSW20], one explicitly requires zero-knowledge proofs [GHL⁺22].

1.1 Previous Work

In general one can distinguish between *tailor-made* approaches and *generic* (or *general-purpose*) approaches to prove knowledge of isogenies in zero-knowledge. Subsequently, we are only focusing on work directly relevant to our approach, and we refer the reader to a recent comprehensive survey of proofs of knowledge of isogenies by Beullens et al. [BDFGP23] for a complete overview.

For tailor-made approaches, the most recent work is the one of Basso et al. [BCC⁺23], which builds on the SIDH proof of knowledge from [DFJP14, DDGZ22] and achieves statistical zero-knowledge. One main limitation of this (and most previous approaches, with the exception of [DKL⁺20]) is that the small challenge space requires numerous parallel executions of the protocol in order to reduce the soundness error. Moreover, the knowledge soundness achieved in [BCC⁺23] is not exact but only relaxed, i.e. while the relation is intended to prove knowledge of a d -isogeny, one can only extract an $\ell^{2i}d$ -isogeny for some small prime ℓ and $0 \leq i \leq n$.

The second approach is to take a general-purpose (non-interactive) zero-knowledge proof system that is capable of proving any language in NP, such as a zk-SNARK, and prove the respective isogeny relation using this proof system. While tailor-made approaches might intuitively seem to be more efficient than such a generic approach, there has been enormous progress in the field of zk-SNARKs over the last decade (cf. [Tha22] for a good overview). This has led Cong, Lai and Levin [CLL23] (CLL henceforth) to look into how well such an approach can perform when concretely instantiated with various recent general-purpose zero-knowledge proof systems. As a starting point CLL take the work by Chavez-Saab, Rodríguez-Henríquez and Tibouchi [CSRT22], which constructs isogeny-based verifiable delay functions (VDFs) [BBBF18] using a succinct non-interactive argument (SNARG) system. For the evaluation of their VDF they require to prove isogeny walks between supersingular elliptic curves.

¹ This is a technique often used to avoid a trusted setup for generating the structured reference string (SRS) for succinct non-interactive argument of knowledge systems (zk-SNARKs) [GKM⁺18], and can be seen as a variant where one uses explicit zero-knowledge proofs for the updates [AGRS24] instead of knowledge assumptions [GKM⁺18].

In brief, for a small prime ℓ they consider the supersingular isogeny graph of ℓ -isogenous supersingular elliptic curves (represented by their j -invariants) and want to prove a walk in this graph. Their idea now is to consider the ℓ^{th} modular polynomial $\Phi_\ell(X, Y) \in \mathbb{Z}[X, Y]$, for which it holds that two curves E and E' are ℓ -isogenous (over an algebraic closure) if and only if their j -invariants satisfy $\Phi_\ell(j(E), j(E')) = 0$. Consequently, when aiming to prove a walk in the ℓ -isogeny graph from some starting curve E to some curve E' , we can efficiently represent it as a sequence of successive steps, i.e. a sequence of j -invariants $j_0, j_1, \dots, j_k$ such that $\Phi_\ell(j_i, j_{i+1}) = 0$ for all $i \in \{0, \dots, k-1\}$, and $j_0 = j(E)$ and $j_k = j(E')$. This means that the relation²

$$\mathcal{R}_{\ell^k\text{-ISOPATH}} = \{((E, E'), \phi) \mid \phi: E \rightarrow E' \text{ is an isogeny of degree } \ell^k\}$$

can equivalently be represented by means of the relation

$$\mathcal{R}_{\ell^k\text{-MODPOLY}} = \left\{ \left((E, E'), (j_i)_{i \in [k-1]} \right) \mid \begin{array}{l} \Phi_\ell(j(E), j_1) = 0 \wedge \\ \Phi_\ell(j_{k-1}, j(E')) = 0 \\ \bigwedge_{i \in [k-2]} \Phi_\ell(j_i, j_{i+1}) = 0 \end{array} \right\}. \quad (1)$$

We note that in [CSRT22] the authors do not require the zero-knowledge and knowledge soundness properties for the VDF application, and thus a SNARG suffices. CLL then use the above relation for the specific case of an isogeny of degree 2^k to construct a rank-1 constraint system (R1CS), which is a very popular arithmetization method in state-of-the-art zk-SNARKs. They then take a number of existing plausibly post-quantum zero-knowledge argument systems and in particular Aurora [BCR⁺19], Ligerio [AHIV17] and Limbo [DOT21], which do not need to make additional structured cryptographic assumptions (e.g. such as lattice-based proof systems for R1CS [NS22, BS23]).

Although CLL focus on $\ell = 2$, it is not uncommon for isogeny-based protocols to involve, or at least allow for, other small primes. For instance, the KEM presented in [Mor23] makes use of a 3-smooth isogeny as a public key, and a 5- or 7-smooth isogeny for encapsulation. Similarly, a 3-smooth isogeny is used for the encryption in the PKE schemes from [DFV24] and [BM25, Algorithm 4]. More generally, while the choice $\ell = 2$ is usually done for simplicity, considering different small primes can provide greater flexibility and also allows for trade-offs in the efficiency between the building blocks and the isogeny proofs of knowledge.

The results in [CLL23] show that the efficiency of this general-purpose approach when compared with the recent tailor-made approach in [BCC⁺23] achieves an order of magnitude improvement over proof and verification times, with slightly worse but still comparable proof sizes. Moreover, compared to existing tailor-made solutions, this approach provides a stronger notion of soundness, i.e. an exact instead of a relaxed one.

In this work we ask whether this is the best we can do when targeting R1CS and whether the approach can be generalized to prove the knowledge of isogenies of degree ℓ^k for primes ℓ greater than 2.

1.2 Our Contributions

The goal of this paper is to improve on the state-of-the-art results of [CLL23] for proving the knowledge of an isogeny, and we make the following contributions.

Use of canonical modular polynomials. We consider canonical modular polynomials in place of the classical modular polynomials used in [CSRT22, CLL23], and we show that constructing a proof of knowledge of the corresponding relation $\mathcal{R}_{\ell^k\text{-MODROOT}}$ is computationally equivalent to proving the relations $\mathcal{R}_{\ell^k\text{-ISOPATH}}$ and $\mathcal{R}_{\ell^k\text{-MODPOLY}}$ mentioned above. While the approach via classical modular polynomials stems directly from well-known theoretical results, the same results are not as readily available for the case of canonical polynomials. Therefore we incorporate them and prove connections to the classical modular polynomials via novel use of resultant theory. We also spot a few gaps in the relevant literature and provide new proofs of some basic properties of the modular polynomials over finite fields. For example, we analyze the existence of edges of multiplicity at least three in the supersingular ℓ -isogeny graph for small ℓ – which seems to be known to experts of the field, at least in a weaker form – and we study the relationship between multiple edges in the ℓ -isogeny graph on the one hand and multiple roots of the ℓ^{th} canonical modular polynomial on the other hand. Hence this part might also be of interest beyond the concrete application in this paper.

² In $\mathcal{R}_{\ell^k\text{-ISOPATH}}$ we consider all ℓ^k -isogenies defined over a fixed algebraic closure.

Improved and generalized isogeny proofs of knowledge. By moving to canonical modular polynomials we obtain a more efficient arithmetization for the equivalent relation $\mathcal{R}_{\ell^k\text{-MODPOLY}}$. Moreover, while [CLL23] only consider isogenies of 2-power degrees, we generalize the approach to cover isogenies of degree ℓ^k , where $\ell \in \{2, 3, 5, 7, 13\}$. This is of interest not only because such primes are used in some recent isogeny-based constructions [Mor23, DFV24, BM25], but also because here we reduce the number of constraints further, potentially yielding even more efficient proof systems.

We first encode our new relation into an R1CS over the field $\mathbb{F}_{p^2}$, and subsequently lift these arithmetizations to $\mathbb{F}_p \times \mathbb{F}_p$ in order to obtain a formulation that works over $\mathbb{F}_p$. We are able to exploit the structured nature of the canonical modular polynomials to optimize the resulting arithmetizations. We describe several techniques to minimize the non-zero entries in the constraint matrices when lifting, such as a basis change for product relations and a change of variables for linear relations, which may be applicable more broadly. To give some intuition, this translates to a concrete reduction in prover times from roughly 1.2 seconds to 700 ms and in verifier time from roughly 100 ms to 60 ms for the identification scheme over $\mathbb{F}_p$ described in [CLL23]. Additionally, we provide a circuit for backtracking prevention that is three times more efficient compared to [CLL23]. All these improvements are presented in Section 5.

As multiple of our proofs contain involved calculations, we verify the necessary computational claims with SageMath scripts that can be found in the accompanying GitHub repository³.

1.3 High-level Overview

To improve upon the previous approach of [CLL23], we introduce the canonical modular polynomials $\Phi_\ell^c(X, j)$ and study how, for $\ell \in \{2, 3, 5, 7, 13\}$ and a fixed j -invariant, their roots relate to ℓ -isogenies: First, we use resultant theory to arrive at the **Multiplicity Theorem**, which states that the number of ℓ -isogenies $j_0 \rightarrow j_1$ agrees with the number of solutions of the equation system

$$\Phi_\ell^c(X, j_0) = 0 = \Phi_\ell^c(\ell^s / X, j_1), \quad (2)$$

with the integer exponent $s = 12/(\ell - 1)$. Next, we argue why, in most cases, there are at most two such solutions/isogenies, which allows for efficient computation of these roots via the Euclidean algorithm. Then, except at the j -invariants 0 and 1728, the connection between the solutions and ℓ -isogenies is made more explicit in the **Reconstruction Theorem** by systematically associating to a solution of the above system (2) a kernel polynomial describing an ℓ -isogeny $j_0 \rightarrow j_1$. As we can recover the solution from the coefficients of said kernel polynomial, we use this result as our main tool to show that the solutions lie in $\mathbb{F}_{p^2}$ when considering supersingular j -invariants. Finally, we demonstrate how the developed theory can be leveraged to yield an efficient arithmetization by encoding (a slightly modified variant of) the equation system (2) as an R1CS over both $\mathbb{F}_{p^2}$ and $\mathbb{F}_p$, where we optimize the number of variables, constraints and non-zero entries.

1.4 Concurrent Work

In [LP24], Levin and Pedersen examine radical isogenies and develop a verifiable random function (VRF) from an efficient proof of knowledge of an isogeny. Although the theory behind the two approaches is quite different, they originally obtained a different arithmetization that used the same number of constraints and number of variables asymptotically for $\ell = 2$, with the additional benefit that their approach prevents backtracking for free. However, in an updated version appearing at Asiacrypt 2025, they managed to further reduce the numbers of constraints and variables via a clever substitution, resulting in a system with fewer constraints and variables for $\ell = 2$. In contrast, our approach is more general in two respects: We do not put restrictions on the prime p while they need $p \equiv 3 \pmod{4}$, and we also generalize to isogeny degrees beyond $\ell = 2$, which allows us to further reduce the numbers of constraints and variables of our approach. Regarding backtracking prevention, we improve on the technique of [CLL23, Appendix A] in Subsection 4.4, but nonetheless require one additional constraint per step over both $\mathbb{F}_{p^2}$ and $\mathbb{F}_p$.

³ <https://github.com/QuSAC/IsogenyPoKviaCanonicalModPolys>

2 Preliminaries

2.1 Isogeny Graphs and Classical Modular Polynomials

Let K be a perfect field, and let E_0, E_1 be elliptic curves over K . An *isogeny* is a morphism of curves $\phi: E_0 \rightarrow E_1$ which induces also a surjective group homomorphism on the sets of $\bar{K}$ -rational points. An isogeny of degree n is also called an *n-isogeny*, and two elliptic curves E_0 and E_1 over K are called *n-isogenous* if there exists an n -isogeny $\phi: E_0 \rightarrow E_1$. An isogeny of degree 1 is called an *isomorphism*, an isogeny $\phi: E \rightarrow E$ is called an *endomorphism*, and an endomorphism of degree 1 is called an *automorphism*. We refer the interested reader to Appendix A.1 for other standard definitions, properties and references on elliptic curves and isogenies.

We will say that two n -isogenies $\phi_1: E \rightarrow E_1$ and $\phi_2: E \rightarrow E_2$ are *equivalent* if they are the same up to post-composition with an isomorphism, i.e. if there exists an isomorphism $\sigma: E_1 \rightarrow E_2$ such that $\phi_2 = \sigma \circ \phi_1$. The kernel of an isogeny ϕ can be represented by its *kernel polynomial* [Koh96, §2.4], which is the square-free monic polynomial whose roots are precisely the x -coordinates of non-trivial points in the kernel of ϕ . Closely related to kernel polynomials is the n^{th} *division polynomial* ψ_n of an elliptic curve E (which we scale by $2y$ for even n compared to the usual definition [Was08, p. 81], so that it is always a polynomial in x only): It is the kernel polynomial of the multiplication-by- n endomorphism scaled by the factor n (resp. $2n$) for odd n (resp. even n) as long as the characteristic $\text{char}(K)$ does not divide n .

To each elliptic curve E defined over a field K one can attach a *j-invariant* $j(E) \in K$ which can be computed efficiently from the coefficients of E . Two elliptic curves E_0 and E_1 are isomorphic if and only if $j(E_0) = j(E_1)$, and any $j_0 \in \bar{K}$ is the *j-invariant* of an elliptic curve defined over $K(j_0)$ [Sil09, Proposition III.1.4]. By a slight abuse of terminology, we will often refer to the *number of non-equivalent n-isogenies* $j_0 \rightarrow j_1$ to indicate the number of equivalence classes of n -isogenies starting from a fixed elliptic curve E_0 of *j-invariant* j_0 and landing on some elliptic curve of *j-invariant* j_1 ; note that the number of such equivalence classes does not depend on the choice of representative E_0 of j_0 .

Now fix a prime p and let $q = p^k$ for some positive integer k . An elliptic curve E over $\mathbb{F}_q$ is called *supersingular* if the cardinality of $E(\mathbb{F}_q)$ is congruent to 1 modulo p [Was08, Proposition 4.31]. Given a prime $\ell \neq p$, the *supersingular ℓ -isogeny graph* $\mathcal{G}_\ell(p)$ is the directed multigraph defined as follows: The vertices of $\mathcal{G}_\ell(p)$ shall be the isomorphism classes of supersingular elliptic curves, which we parametrize by their *j-invariants* in $\mathbb{F}_{p^2}$ [Sil09, Theorem V.3.1], and the number of edges $j_0 \rightarrow j_1$ is precisely the number of non-equivalent ℓ -isogenies $j_0 \rightarrow j_1$.

The graph $\mathcal{G}_\ell(p)$ is connected, $(\ell + 1)$ -regular, and it is Ramanujan [Piz90, CL24]. Furthermore, since every isogeny admits a dual isogeny [Sil09, Theorem III.6.1-2], it can almost be considered as an undirected graph; however, curves of *j-invariants* 0 or 1728 have special automorphisms [Sil09, Theorem III.10.1], which can cause asymmetries in the graph for $p \geq 5$: If we write $\mu(0) := 3$, $\mu(1728) := 2$ and $\mu(j) := 1$ for $j \notin \{0, 1728\}$, then there are $\mu(j_0)/\mu(j_1)$ times as many non-equivalent ℓ -isogenies $j_0 \rightarrow j_1$ as there are non-equivalent ℓ -isogenies $j_1 \rightarrow j_0$ (cf. [AAM19, Formula (11)]).

In this paper we will consider random walks on $\mathcal{G}_\ell(p)$, i.e. sequences

$$j(E_0) \rightarrow j(E_1) \rightarrow \dots \rightarrow j(E_k)$$

of adjacent *j-invariants* in $\mathcal{G}_\ell(p)$. It is easy to check whether two given *j-invariants* belong to a pair of ℓ -isogenous elliptic curves. To this end, one can use the so-called *classical modular polynomials* (see [Mü95, §4.3], [AAM19, §2.4] and [Sut13, §2.3]): The ℓ^{th} classical modular polynomial $\Phi_\ell(X, Y)$ is a bivariate polynomial with integer coefficients whose roots are given by the pairs of *j-invariants* of ℓ -isogenous elliptic curves – more precisely, given the prime ℓ and any two elliptic curves E, E' over a field K with $\text{char}(K) \neq \ell$, the number of non-equivalent ℓ -isogenies $E \rightarrow E'$ is equal to the multiplicity of $j(E')$ as a root of $\Phi_\ell(j(E), Y)$.

2.2 Resultants

Let R be an integral domain and let $g, h \in R[X]$ be non-zero polynomials. The *Sylvester matrix* of g and h and especially its determinant, the *resultant* $\text{res}(g, h)$ of g and h , are important algebraic tools to detect common divisors between two polynomials. We only state the necessary properties here and give the theoretical background together with proofs for the below results in Appendix B.

Proposition 1. Let R be an integral domain, let $g, h \in R[X]$ be non-zero polynomials and let $\varphi: R \rightarrow S$ be a ring homomorphism of integral domains, extended to a ring homomorphism $\varphi: R[X] \rightarrow S[X]$ via coefficient-wise application. Then the following holds:

(a) If φ preserves the degrees of g and h , then

$$\text{res}(\varphi(g), \varphi(h)) = \varphi(\text{res}(g, h)).$$

(b) $\text{res}(g, h) = 0$ if and only if g and h share a common divisor of positive degree.

Proof. Lemma 16 and Corollary 17. □

Remark 1. We note that, as the *discriminant* of g is defined as the resultant of g and $\frac{\partial}{\partial X}g$ up to scaling, the above results also translate to the discriminant; see Corollary 22 for a precise formulation of the first claim for discriminants.

Proposition 2. Let $R = K[Y]$ be a polynomial ring over a field K and fix an element $y_0 \in K$. Additionally let $g, h \in R[X]$ be non-zero polynomials and extend the K -linear evaluation homomorphism $\varphi: R \rightarrow K$ given by $Y \mapsto y_0$ to a ring homomorphism $\varphi: R[X] \rightarrow K[X]$ via coefficient-wise application. Further suppose that φ preserves the X -degrees of g and h , and write

$$m := \deg \gcd(\varphi(g), \varphi(h)).$$

Then

$$\left. \frac{\partial^k}{\partial Y^k} \right|_{Y=y_0} \text{res}_X(g, h) = 0 \text{ for } k \in \{0, \dots, m-1\}.$$

Proof. Corollary 19. □

2.3 Zero-Knowledge Argument Systems and R1CS

As stated before, in this work we will use generic techniques to prove the knowledge of isogenies, improving and expanding on the previous results of [CLL23]. Since our arithmetization is broadly applicable across different argument systems and we do not need any formal properties of an argument system throughout this work, we will omit a full formal treatment of zk-SNARKS. For a comprehensive formal treatment, readers are referred to the respective proof systems [BCR⁺19, AHIV17, XZZ⁺19].

A zk-SNARK is a non-interactive argument system that is complete, knowledge-sound, zero-knowledge and succinct. The proving algorithm takes a statement and witness pair (s, w) for some NP-relation and generates a proof π . There is a verification algorithm to check the validity of a proof. Completeness indicates that a valid proof can be generated from any pair (s, w) in the relation. Knowledge soundness means that any prover that can generate a valid proof for a statement s needs to know a corresponding witness w . Zero-knowledge means that the proof does not reveal any information about the witness. An argument system is succinct if the proof size is small and the proof can be verified efficiently. Usually, both proof size and verifier time are required to be polynomial in $|x|$ and polylogarithmic in $|w|$.

In this paper we design an efficient arithmetization in the form of a rank-1 constraint system (R1CS). This represents a popular choice of arithmetization and it allows us to cover many different proof systems. An R1CS is defined as follows:

Definition 1 ([BCR⁺19]). The relation $\mathcal{R}_{\text{R1CS}}$ is the set of pairs $((\mathbb{F}, k, n, m, A, B, C, v), w)$ where $\mathbb{F}$ is a finite field, $k, n, m \in \mathbb{N}$ denote the numbers of inputs, variables and constraints respectively ($k \leq n$), A, B, C are $m \times (1+n)$ -matrices over $\mathbb{F}$, $v \in \mathbb{F}^k$, and $w \in \mathbb{F}^{n-k}$, such that for all $i \in [m]$

$$\left(\sum_{j=0}^n A_{ij} z_j \right) \cdot \left(\sum_{j=0}^n B_{ij} z_j \right) = \left(\sum_{j=0}^n C_{ij} z_j \right),$$

where $(1, v, w) =: z = (z_j)_j \in \mathbb{F}^{1+n}$.

It is worth noting that the efficiency of proving and verifying knowledge of a witness may depend on different aspects of the R1CS, depending on the proof system that is used. For example, the prover time of pairing-based SNARKs is usually $O(n)$, the proof size is constant, and the verifier time is $O(k)$ [Gro16, Lip22]. On the other hand, [BCR⁺19, AHIV17, XZS22] have prover time proportional to the circuit size, which corresponds to the number of non-zero entries in the constraint matrices A , B and C , which we will denote by nnz . Lastly, for [DOT21] the proof size and prover and verifier times seem to be determined by the number of multiplications, corresponding to the number of R1CS constraints m . The efficiency of the arithmetization in [CLL23] is only quantified through the number of constraints m and optimized using this metric.

In this work, we will provide all of n , m and nnz . When optimizing, we will focus on the latter two, since the number of variables is mostly relevant for the non-post-quantum secure pairing-based SNARKs. Often, optimizing for one metric also improves another, such as when a linear constraint can be removed to eliminate a variable, but this is not always the case. As we will see in Section 5, we achieve very efficient constraint systems in terms of all three metrics.

3 Canonical Modular Polynomials

The classical modular polynomials tend to have many non-vanishing coefficients, which makes these polynomials quite expensive to handle in an R1CS. To be more precise, the polynomial $\Phi_\ell(X, Y)$ is symmetric in X and Y , of degree $\ell + 1$ in both variables [Lan87, Theorem 5.2.3], and typically most of the possible mixed monomials $X^i Y^j$ with $i, j \leq \ell$ occur. For example, the third classical modular polynomial is given by (see [CFA⁺06, Example 17.18])

$$\begin{aligned}\Phi_3(X, Y) = & -X^3 Y^3 + X^4 + Y^4 + 2232(X^3 Y^2 + X^2 Y^3) \\ & - 1069956(X^3 Y + X Y^3) + 36864000(X^3 + Y^3) \\ & + 2587918086X^2 Y^2 + 8900222976000(X^2 Y + X Y^2) \\ & + 452984832000000(X^2 + Y^2) - 770845966336000000XY \\ & + 1855425871872000000000(X + Y).\end{aligned}$$

Luckily, there is a related class of polynomials called *canonical modular polynomials* Φ_ℓ^c [Ler97, §3.3.2][Mor95, §2.2][Mü95, §5.1-2], which are asymmetric and have a smaller degree in the second variable. To contrast our previous example, the third canonical modular polynomial is given by

$$\Phi_3^c(X, j) = X^4 + 36X^3 + 270X^2 + 756X + 729 - X \cdot j.$$

The general (modular) construction. In general, the ℓ^{th} canonical modular polynomial is constructed as follows: Letting $s = 12 / \gcd(12, \ell - 1)$ denote the smallest non-zero natural number such that

$$\kappa := \frac{s \cdot (\ell - 1)}{12}$$

is an integer, and letting η denote the Dedekind η function, the function

$$f(\tau) := \ell^s \left(\frac{\eta(\ell\tau)}{\eta(\tau)} \right)^{2s}$$

is a modular function of weight 0 that is invariant under Möbius transformations of τ given by elements of

$$\Gamma_0(\ell) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z}) : c \equiv 0 \pmod{\ell} \right\}.$$

The finitely many cosets of $\text{SL}_2(\mathbb{Z}) / \Gamma_0(\ell)$ are represented by the matrices

$$S_n = \begin{pmatrix} 0 & -1 \\ 1 & n \end{pmatrix}, 0 \leq n < \ell, \text{ and } S_\ell = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and one defines the canonical modular polynomial as

$$\Phi_\ell^c(X) := \prod_{n=0}^{\ell} (X - f(S_n(\tau))) = (X - f(\tau)) \prod_{n=0}^{\ell-1} \left(X - f\left(\frac{-1}{\tau + n}\right) \right).$$

This polynomial can now be expressed as a polynomial $\Phi_\ell^c(X, j)$ with integer coefficients in X and the j -invariant $j = j(\tau)$, which itself is a modular function of weight 0 invariant under transformations given by matrices in $\mathrm{SL}_2(\mathbb{Z})$. Moreover, the degree of $\Phi_\ell^c(X, j)$ in the second variable j is given by κ (see [Mü95, Equations (5.1-2) and Lemma 5.7]).

Restricting to $\ell \in \{2, 3, 5, 7, 13\}$. The ℓ^{th} canonical modular polynomial thus has, for suitable coefficients $c_{i,m} \in \mathbb{Z}$, the form

$$\Phi_\ell^c(X, j) = \sum_{i=0}^{\ell+1} \sum_{m=0}^{\kappa} c_{i,m} X^i j^m.$$

In the case $\kappa = 1$, i.e. $\ell \in \{2, 3, 5, 7, 13\}$, these polynomials are hence quite small – we list them in Appendix A.2, and hereafter we will work with these explicitly given polynomials directly.

Lemma 3. Let $\ell \in \{2, 3, 5, 7, 13\}$. Setting $c_i := c_{i,0} \in \mathbb{Z}$, the ℓ^{th} canonical modular polynomial $\Phi_\ell^c(X, j)$ then is of the form

$$\Phi_\ell^c(X, j) = X^{\ell+1} + \sum_{i=1}^{\ell} c_i X^i + \ell^s - X \cdot j = \Phi_\ell^c(X, 0) - X \cdot j.$$

In particular, the rational function

$$\Theta_\ell^c(X, j) := \Phi_\ell^c(\ell^s/X, j) \cdot X^{\ell+1}/\ell^s = X^{\ell+1} + \sum_{i=1}^{\ell} c_i \ell^{s \cdot (i-1)} X^{\ell+1-i} + \ell^{s \cdot \ell} - X^\ell \cdot j$$

is a polynomial with integer coefficients. With $\mathcal{J}_\ell(X) := \Phi_\ell^c(X, 0)/X$ we furthermore have that for any $f \in K^\times$ and $j_0 \in K$:

$$j_0 = \mathcal{J}_\ell(f) \text{ if and only if } \Phi_\ell^c(f, j_0) = 0.$$

Proof. The claim on the form of $\Phi_\ell^c(X, j)$ follows from direct inspection, and the claim on the form of $\Theta_\ell^c(X, j)$ is an immediate consequence. Furthermore we have $\Phi_\ell^c(X, j) = X \cdot \mathcal{J}_\ell(X) - X \cdot j$, which implies the claim on $\mathcal{J}_\ell(X)$. $\square$

Since the constant coefficients of $\Phi_\ell^c(X, j)$ and $\Theta_\ell^c(X, j)$ are powers of ℓ , we obtain:

Corollary 4. Let $\ell \in \{2, 3, 5, 7, 13\}$, assume $\mathrm{char}(K) \neq \ell$ and let $j_0, j_1 \in K$ be j -invariants with $j_0 \neq j_1$. Then

$$\gcd(\Phi_\ell^c(X, j_0), \Phi_\ell^c(X, j_1)) = 1 = \gcd(\Theta_\ell^c(X, j_0), \Theta_\ell^c(X, j_1)).$$

Roadmap for this section. In Subsection 3.1 we investigate how, for a given $j_0 \in K$, the roots of $\Phi_\ell^c(X, j_0)$ relate to the number of ℓ -isogenies $j_0 \rightarrow j_1$ to some $j_1 \in K$, culminating in the Multiplicity Theorem. Then we discuss in Subsection 3.2 how many of the related ℓ -isogenies can lead to the same target j_1 (Proposition 8, Theorem 9, Corollary 10) to argue how these roots can be computed efficiently. Finally, we analyze in Subsection 3.3 in which field extension a root f lies by giving a connection to kernel polynomials via the Reconstruction Theorem, which we then exploit together with previous results to determine the splitting behavior of $\Phi_\ell^c(X, j_0)$ for a supersingular j -invariant j_0 in Theorem 11.

In relation to ℓ -isogenies, the above mentioned restriction to the five primes $\ell \in \{2, 3, 5, 7, 13\}$ satisfying $\kappa = 1$ also has a conceptual reason. In fact, these primes are precisely those for which the modular curve $X_0(\ell)$ has genus 0 by [Tsu13, Proposition 2.3.5]. In relation to the modular construction, this gives a high-level intuition of why we obtain a compact representation of ℓ -isogenies: The elements of $X_0(\ell)$, which are the edges in $\mathcal{G}_\ell(p)$, can then be parametrized (up to issues at the ‘ramified’ points $j \in \{0, 1728\}$) by $f \in \mathbb{F}_p^\times$. This parametrization, which we will later analyze in the Reconstruction Theorem, has already been studied in the works of Fricke [Fri11, Section 2, Chapters 4-5], Mestre [Mes86, §5], and Elkies [Elk98, §4].

3.1 The Multiplicity Theorem

Unfortunately, it is no longer true that $\Phi_\ell^c(j_0, j_1) = 0$ if j_0 and j_1 are ℓ -isogenous j -invariants in $\overline{\mathbb{F}_p}$. Instead, taking inspiration from the modular interpretation given e.g. by [Ler97, p. 41], we will show that we need to find a common root f of the two functions $\Phi_\ell^c(X, j_0)$ and $\Phi_\ell^c(\ell^s/X, j_1)$, where $s = 12/(\ell - 1)$ is defined as above. To prove a more precise version of this claim, we first relate the classical modular polynomial to the canonical modular polynomial via resultant theory.

Proposition 5. For any $\ell \in \{2, 3, 5, 7, 13\}$ we have, computing over the coefficient ring $R = \mathbb{Z}[J_0, J_1]$, the resultant equation

$$\text{res}_X(\Theta_\ell^c(X, J_1), \Phi_\ell^c(X, J_0)) = \ell^{s \cdot \ell} \cdot \Phi_\ell(J_0, J_1).$$

In particular, suppose that we have a field K of characteristic $\text{char}(K) \notin [\ell]$ as well as $j_0, j_1 \in K$ with $m := \deg \gcd(\Phi_\ell^c(X, j_0), \Theta_\ell^c(X, j_1))$. Then

$$\left. \frac{\partial^k}{\partial J_1^k} \right|_{J_1=j_1} \Phi_\ell(j_0, J_1) = 0 \text{ for } k \in \{0, \dots, m-1\},$$

i.e. there are at least m non-equivalent ℓ -isogenies from j_0 to j_1 .

Proof. The first claim is checked in the script `additional_computations.sage`. For the second claim we first apply the ring homomorphism $\mathbb{Z}[J_0] \rightarrow K$ defined by $J_0 \mapsto j_0$ in view of Proposition 1(a) to obtain in $K[J_1]$ the equality

$$\text{res}_X(\Theta_\ell^c(X, J_1), \Phi_\ell^c(X, j_0)) = \ell^{s \cdot \ell} \cdot \Phi_\ell(j_0, J_1).$$

Now we consider the K -linear evaluation homomorphism $\varphi: K[J_1] \rightarrow K$ given by $J_1 \mapsto j_1$, extended via coefficient-wise application to $\varphi: K[J_1][X] \rightarrow K[X]$. As φ preserves the X -degrees of $g := \Theta_\ell^c(X, J_1)$ and $h := \Phi_\ell^c(X, j_0)$, we are exactly in the situation of Proposition 2. With $\text{char}(K) \notin [\ell]$ we thus deduce that $\text{res}_X(g, h) = \ell^{s \cdot \ell} \cdot \Phi_\ell(j_0, J_1)$ has a root of multiplicity at least m at j_1 , and $\text{char}(K) \neq \ell$ yields the claim. $\square$

The previous relation will be the main tool in establishing the desired connection between the classical and the canonical modular polynomial. For the proof we need to analyze root multiplicities of the canonical modular polynomial in the next two results, which can also be found in [Tsu13, §4.3] in the language of modular curves.

Lemma 6. Let $\ell \in \{2, 3, 5, 7, 13\}$, assume $\text{char}(K) \neq \ell$ and let $j_0 \in K$. Then $\Phi_\ell^c(X, j_0)$ has a double root in $\overline{K}$ if and only if $j_0 = 0$ or $j_0 = 1728$.

Proof. To simplify notation we may assume that K is algebraically closed. In view of Lemma 3 we obtain the univariate polynomial

$$\mathcal{D}_\ell(X) := j + \frac{\partial}{\partial X} \Phi_\ell^c(X, j).$$

Now an element $f \in K$ is a double root of $\Phi_\ell^c(X, j_0)$ if and only if it is non-zero (due to $\text{char}(K) \neq \ell$) and satisfies $\mathcal{J}_\ell(f) = j_0 = \mathcal{D}_\ell(f)$. From this we see that the double roots are precisely the common roots of $\Phi_\ell^c(X, j_0)$ and the polynomial

$$\mathcal{D}_\ell(X) \cdot X - \mathcal{J}_\ell(X) \cdot X = -\Phi_\ell^c(X, \mathcal{D}_\ell(X)),$$

which has leading coefficient ℓ . In the script `additional_computations.sage` we check that there are $e, m, n \in \mathbb{N}$ such that

$$\text{res}_X(\Phi_\ell^c(X, J), -\Phi_\ell^c(X, \mathcal{D}_\ell(X))) = (-1)^\ell \cdot \ell^e \cdot (J - 0)^m \cdot (J - 1728)^n,$$

computed over the coefficient ring $R = \mathbb{Z}[J]$. Therefore $\text{char}(K) \neq \ell$ allows us to apply Proposition 1 (with the homomorphism $\mathbb{Z}[J] \rightarrow K$ given by the evaluation $J \mapsto j_0$) to deduce the claimed equivalence. $\square$

The second result discusses the special j -invariants 0 and 1728; for an explicit list of the factors given below we refer the reader to Appendix A.3.

Lemma 7. For each $\ell \in \{2, 3, 5, 7, 13\}$ there are (ℓ -dependent) monic polynomials g_0 and g_{1728} in $\mathbb{Z}[X]$ of degree at most 2 and (ℓ -dependent) monic non-constant polynomials $h_{0,\pm}$ and $h_{1728,\pm}$ in $\mathbb{Z}[X]$ such that

$$\Phi_\ell^c(X, 0) = g_0 \cdot h_{0,+}^3, \quad \Theta_\ell^c(X, 0) = g_0 \cdot h_{0,-}^3$$

and

$$\Phi_\ell^c(X, 1728) = g_{1728} \cdot h_{1728,+}^2, \quad \Theta_\ell^c(X, 1728) = g_{1728} \cdot h_{1728,-}^2.$$

Moreover, if K is a field with $\text{char}(K) \notin \{2, 3, \ell\}$ and $j^* \in \{0, 1728\}$, then each $h_{j^*,\pm}$ has only simple roots, and does not share any roots with g_{j^*} in $\bar{K}$.

Proof. In the SageMath script `additional_computations.sage` we confirm the above factorizations as well as that, for $j^* \in \{0, 1728\}$, the prime factors of $\deg(h_{j^*,\pm})$ lie in $\{2, 3\}$, and that the prime factors of $\text{disc}(h_{j^*,\pm})$ and $\text{res}(g_{j^*}, h_{j^*,\pm})$ lie in $\{2, 3, \ell\}$. To prove the additional claim, we now make use of resultant theory once more by considering the unique homomorphism $\varphi: \mathbb{Z} \rightarrow K$. This homomorphism preserves the degree of $h_{j^*,\pm}$ and, as the prime factors of $\deg(h_{j^*,\pm})$ are contained in $\{2, 3\}$, the degree of $\frac{\partial}{\partial X} h_{j^*,\pm}$ due to our assumption on $\text{char}(K)$. Therefore this assumption and Proposition 1(a) (cf. Remark 1) yield

$$\text{disc}(\varphi(h_{j^*,\pm})) = \varphi(\text{disc}(h_{j^*,\pm})) \neq 0,$$

so $h_{j^*,\pm}$ cannot have a double root in $\bar{K}$ by Proposition 1(b). Similarly we obtain

$$\text{res}(\varphi(g_{j^*}), \varphi(h_{j^*,\pm})) = \varphi(\text{res}(g_{j^*}, h_{j^*,\pm})) \neq 0,$$

i.e. g_{j^*} and $h_{j^*,\pm}$ cannot have a common root in $\bar{K}$ by Proposition 1(b). $\square$

With the above preparations we are finally ready to state and prove the following crucial relation between the classical and canonical modular polynomial:

Multiplicity Theorem. Let $\ell \in \{2, 3, 5, 7, 13\}$, let K be a field of characteristic $\text{char}(K) \notin \{2, \dots, \ell\}$ and let $j_0, j_1 \in K$. Then there are exactly as many non-equivalent ℓ -isogenies $j_0 \rightarrow j_1$ as there are roots $f \in \bar{K}^\times$ of $\Phi_\ell^c(X, j_0)$ (counted with multiplicity) such that $\Phi_\ell^c(\ell^s / f, j_1) = 0$. In particular, j_0 and j_1 are ℓ -isogenous if and only if there exists an $f \in \bar{K}^\times$ such that

$$\Phi_\ell^c(f, j_0) = 0 = \Phi_\ell^c(\ell^s / f, j_1). \quad (\text{MT})$$

Proof. To simplify notation we assume that K is algebraically closed. As before we consider the polynomial $\Theta_\ell^c(X, j_1) = \Phi_\ell^c(\ell^s / X, j_1) \cdot X^{\ell+1} / \ell^s$ instead of the rational function $\Phi_\ell^c(\ell^s / X, j_1)$, noting that it has the same roots due to $\text{char}(K) \neq \ell$. For any $j_0, j_1 \in K$ we write $\beta_\ell(j_0, j_1)$ for the number of roots of $\Phi_\ell^c(X, j_0)$ (counted with multiplicity) that are roots of $\Theta_\ell^c(X, j_1)$, and we write $\nu_\ell(j_0, j_1)$ for the number of non-equivalent ℓ -isogenies from j_0 to j_1 .

Hence our goal is to show that $\beta_\ell(j_0, j_1) = \nu_\ell(j_0, j_1)$; however, it suffices to prove the inequality $\beta_\ell(j_0, j_1) \leq \nu_\ell(j_0, j_1)$ for all $j_0, j_1 \in K$. Indeed, summing both quantities over all possible j_1 for a fixed $j_0 \in K$ then yields

$$\ell + 1 = \deg_X(\Phi_\ell^c(X, j_0)) = \sum_{j_1 \in K} \beta_\ell(j_0, j_1) \leq \sum_{j_1 \in K} \nu_\ell(j_0, j_1) = \ell + 1$$

in view of Lemma 3, and thus all inequalities have to be equalities.

To prove the inequalities, we first note that Proposition 5 immediately yields

$$\deg \gcd(\Phi_\ell^c(X, j_0), \Theta_\ell^c(X, j_1)) \leq \nu_\ell(j_0, j_1). \quad (3)$$

We will use this inequality in the following, but we need to work through a slightly tedious case distinction: First, if $j_0 \notin \{0, 1728\}$, then any root of $\Phi_\ell^c(X, j_0)$ is a simple root by Lemma 6, so

$$\beta_\ell(j_0, j_1) = \deg \gcd(\Phi_\ell^c(X, j_0), \Theta_\ell^c(X, j_1)) \leq \nu_\ell(j_0, j_1).$$

Next we consider an edge case: Let $\text{char}(K) = 3$ and let $j_0 = j^* = 0 = 1728$ be the unique supersingular j -invariant in K (cf. [Sil09, §V.4]). Then all $\ell + 1$ non-equivalent ℓ -isogenies starting from j^* are loops, and one can directly verify that we have $\Theta_\ell^c(X, 0) = \Phi_\ell^c(X, 0)$ over K .

Therefore, for $j_1 \neq j^*$, Corollary 4 yields

$$\beta_\ell(j^*, j^*) = \ell + 1 = v_\ell(j^*, j^*) \quad \text{and} \quad \beta_\ell(j^*, j_1) = 0 = v_\ell(j^*, j_1);$$

thus we may now assume $\text{char}(K) \notin \{2, 3, \ell\}$. To proceed we recall the multiplicity factors

$$\mu(0) = 3, \quad \mu(1728) = 2 \quad \text{and} \quad \mu(j) = 1$$

for $j \in K \setminus \{0, 1728\}$, and consider a special j -invariant $j_0 = j^* \in \{0, 1728\}$. For $j^* \neq j_1$ any (distinct) root f of $\Phi_\ell^c(X, j^*)$ such that $\Theta_\ell^c(f, j_1) = 0$ then has multiplicity $\mu(j^*)$ (resp. $\mu(j_1)$) as a root of $\Phi_\ell^c(X, j^*)$ (resp. of $\Theta_\ell^c(X, j_1)$). Indeed, if $j_1 \notin \{0, 1728\}$ the second claim follows from Lemma 6, and otherwise all multiplicities are derived from Lemma 7 since Corollary 4 and the assumption $j^* \neq j_1$ force f to be a root of $h_{j^*,+}$ and $h_{j_1,-}$ (if $j_1 \in \{0, 1728\}$), which only have simple roots by our restriction on the characteristic.

With the inequality (3) and $\mu(1728) < \mu(0)$ we hence see that the roots of the greatest common divisor of $\Phi_\ell^c(X, 1728)$ and $\Theta_\ell^c(X, 0)$ all have multiplicity $\mu(1728)$, so we obtain

$$\beta_\ell(1728, 0) = \deg \gcd(\Phi_\ell^c(X, 1728), \Theta_\ell^c(X, 0)) \leq v_\ell(1728, 0).$$

Using the multiplicity-preserving correspondence $f \mapsto \ell^s / f$ between roots of $\Phi_\ell^c(X, j)$ and roots of $\Theta_\ell^c(X, j)$, we further deduce from the above analysis that

$$\beta_\ell(j^*, j_1) = \frac{\mu(j^*)}{\mu(j_1)} \cdot \beta_\ell(j_1, j^*) \leq \frac{\mu(j^*)}{\mu(j_1)} \cdot v_\ell(j_1, j^*) = v_\ell(j^*, j_1)$$

in all other cases with $j^* \neq j_1$, where the middle inequality has been derived in previous cases and the last equality is due to the larger automorphism groups at the special j -invariants 0 and 1728, as explained in Subsection 2.1.

The final case to consider is $j^* = j_1$, where Lemma 7 also does the heavy lifting: Here any root f of $\Phi_\ell^c(X, j^*)$ is either a root of g_{j^*} – then with the same multiplicity for both $\Phi_\ell^c(X, j^*)$ and $\Theta_\ell^c(X, j^*)$ – or it is a root of both $h_{j^*,\pm}$, in which case its multiplicity for both $\Phi_\ell^c(X, j^*)$ and $\Theta_\ell^c(X, j^*)$ is $\mu(j^*)$ since each $h_{j^*,\pm}$ only has simple roots. Therefore we also deduce

$$\beta_\ell(j^*, j^*) = \deg \gcd(\Phi_\ell^c(X, j^*), \Theta_\ell^c(X, j^*)) \leq v_\ell(j^*, j^*)$$

from the inequality (3), and this finishes the proof. $\square$

Remark 2. In fact, it is true for all primes $\ell \in \mathbb{N}$ and ℓ -isogenous j -invariants $j_0, j_1 \in \overline{\mathbb{F}_p}$ that we can always find a common root of the system (MT). Indeed, we can first view j_0 and j_1 as reductions modulo p of CM j -invariants $J_0, J_1 \in \mathbb{C}$ ([Lan87, Theorem 13.5.14]) that are integral by [Cox13, Theorem 11.1]. Now equations (5.2-4) in [Mü95, §5] show that there is a common solution $\tilde{f} \in \mathbb{C}$ that has to be integral as it satisfies the polynomial $\Phi_\ell^c(X, J_0)$; therefore it can be reduced to a solution $f \in \overline{\mathbb{F}_p}$ of the system (MT).

However, the restriction to $\kappa = 1$ is crucial for the other direction: For example, for $\ell = 11$, $p = 61$ the j -invariants $j_0 = 41$ and $j_1 = 37$ are not ℓ -isogenous over $\overline{\mathbb{F}_p}$; in fact, they are not even isogenous since j_0 is supersingular, whereas j_1 is ordinary. Nonetheless, either root of the polynomial $X^2 + 3X - 27 \in \mathbb{F}_{p^2}[X]$ gives a solution to the system (MT).

In spite of that, experiments seem to suggest that there are still *at most* as many ℓ -isogenies $j_0 \rightarrow j_1$ as there are roots from j_0 to j_1 (counted as in the Multiplicity Theorem). Note that this does not contradict our previous findings since Corollary 4 fails for $\kappa > 1$: For instance, in our example we have

$$\gcd(\Phi_{11}^c(X, 41), \Phi_{11}^c(X, 50)) = X^2 - 30X - 1 \in \mathbb{F}_p[X].$$

3.2 Isogeny Relations and Root Computation

Recall that our goal is to build an efficient proof of knowledge for isogenies of degree ℓ^k (where $\ell \in \{2, 3, 5, 7, 13\}$), or, equivalently, for the relation $\mathcal{R}_{\ell^k\text{-MODPOLY}}$ (Equation (1)). However, to apply the canonical modular polynomials we instead need to consider the relation

$$\mathcal{R}_{\ell^k\text{-MODROOT}} := \left\{ \left((E, E'), \left((j_i)_{i \in [k-1]}, (f_i)_{i \in [k]} \right) \right) \mid \bigwedge_{i \in [k-1]} \begin{array}{l} \Phi_\ell^c(f_1, j(E)) = 0 \wedge \\ \Theta_\ell^c(f_k, j(E')) = 0 \\ \Theta_\ell^c(f_i, j_i) = 0 = \Phi_\ell^c(f_{i+1}, j_i) \end{array} \right\}.$$

With the Multiplicity Theorem we see that simply omitting the roots $(f_i)_{i \in [k]}$ from the witness brings us back to $\mathcal{R}_{\ell^k\text{-MODPOLY}}$, so this new relation requires additional information from the prover. To gauge the related additional work for the prover, we will investigate two questions: *How many roots can the system (MT) have? And in which field do these roots lie?*

How many roots? To answer the first question in view of the Multiplicity Theorem, we investigate the number of ℓ -isogenies between j -invariants more closely. The following result is a consequence of the well-known structure of *ordinary isogeny volcanoes*:

Proposition 8. Let ℓ be a prime and suppose that we have two j -invariants $j_0, j_1 \in \overline{\mathbb{F}_p}$ for some prime $p \neq \ell$. If j_0 is ordinary, then the following holds:

- (a) If $j_0 \notin \{0, 1728\}$ or $j_0 = j_1$, then there are at most two non-equivalent ℓ -isogenies $j_0 \rightarrow j_1$.
- (b) If $j_0 = 0 \neq j_1$, then there are at most three non-equivalent ℓ -isogenies $j_0 \rightarrow j_1$ and at most one ℓ -isogeny $j_1 \rightarrow j_0$ up to equivalence.
- (c) If $j_0 = 1728 \neq j_1$, then there are at most two non-equivalent ℓ -isogenies $j_0 \rightarrow j_1$ and at most one ℓ -isogeny $j_1 \rightarrow j_0$ up to equivalence.

Proof. This follows immediately from [Sut13, Theorem 7 & Remark 8] (note, however, that for $\ell = 2$ there is exactly 1 vertex at the first level of the ordinary isogeny graph component containing 1728 – the second formula given in Remark 8 only holds for odd ℓ). $\square$

Remark 3. Müller claims in [Mü95, Lemma 4.14] that the $\ell + 1$ non-equivalent ℓ -isogenies defined on an ordinary curve over $\mathbb{F}_p$ with j -invariant not in $\{0, 1728\}$ map to $\ell + 1$ distinct j -invariants, i.e. to $\ell + 1$ non-isomorphic elliptic curves. However, if we consider $p = 29$, $\ell = 7$ and the two j -invariants $j_0 = 23$ and $j_1 = 12$ (noting that $1728 \equiv 17 \pmod{29}$), then the curve

$$E_0: y^2 = x^3 + 21x + 26$$

satisfies $j(E_0) = 23 = j_0$ and admits two 7-isogenies α_1 and α_2 to the elliptic curve

$$E_1: y^2 = x^3 + 6x + 9$$

of j -invariant $j(E_1) = 12 = j_1$.

Importantly, the kernels of these two 7-isogenies are distinct (and hence the isogenies are not equivalent) since their kernel polynomials

$$x^3 + 2x^2 + 21x + 16 \quad \text{and} \quad x^3 + 14x^2 + 13x + 23$$

are distinct. The issue is that the endomorphism $\hat{\alpha}_2 \circ \alpha_1$ of degree 7^2 is not equivalent to the multiplication-by-7 isogeny [7] on E_0 , which again can be checked by comparing the kernel polynomials of these two endomorphisms.

In general, Proposition 8 does not extend to supersingular j -invariants – the following example can, using the Multiplicity Theorem, easily be checked with both the classical and the canonical modular polynomial:

Example 1. For $\ell = 7$ and $p = 71$ there are 6 non-equivalent ℓ -isogenies $0 \rightarrow 48$, 2 non-equivalent ℓ -isogenies $48 \rightarrow 0$ and 4 non-equivalent ℓ -isogenies $40 \rightarrow 40$.

Luckily, we can strictly limit when the claims of Proposition 8 do not transfer to supersingular j -invariants in our setting:

Theorem 9. Let $\ell \leq 13$ be a prime. Then there is a prime $p_\ell < 4\ell^3$ (given in Table 1) such that for any prime $p > p_\ell$ and any two supersingular j -invariants $j_0, j_1 \in \mathbb{F}_{p^2}$ the following holds:

- (a) If $j_0 \notin \{0, 1728\}$ or $j_0 = j_1$, then there are at most two non-equivalent ℓ -isogenies $j_0 \rightarrow j_1$.
- (b) If $j_0 = 0 \neq j_1$, then there are at most three non-equivalent ℓ -isogenies $j_0 \rightarrow j_1$ and at most one ℓ -isogeny $j_1 \rightarrow j_0$ up to equivalence.
- (c) If $j_0 = 1728 \neq j_1$, then there are at most two non-equivalent ℓ -isogenies $j_0 \rightarrow j_1$ and at most one ℓ -isogeny $j_1 \rightarrow j_0$ up to equivalence.

Proof. We first argue that for $j_0 \neq 0$ there are at most two non-equivalent ℓ -isogenies $j_0 \rightarrow j_1$. If this were not the case, then $\Phi_\ell(j_0, Y)$ would have a triple root at $Y = j_1$, and in Appendix B – specifically Proposition 21 – we use resultant theory to computationally check that this can only happen up to the prime $p_\ell < 4\ell^3$ given in Table 1 below.

In the script `additional_computations.sage` we check that $\Phi_\ell(0, 1728)$ does not have a prime factor larger than p_ℓ , i.e. 0 and 1728 cannot be ℓ -isogenous for $p > p_\ell$. Now consider $j_0 = 0$, which forces $p \equiv 2 \pmod 3$ (cf. [Sil09, Example V.4.4]). For $\ell \in \{2, 3\}$ we check in the same script that, due to $p > p_\ell$, $\Phi_\ell(0, Y)$ has a triple root $j_1 \in \mathbb{F}_p \setminus \{0, 1728\}$, and for $\ell = 3$ a single root at 0. Since the number of non-equivalent ℓ -isogenies $0 \rightarrow j_1$ is three times the number of non-equivalent ℓ -isogenies $j_1 \rightarrow 0$, both claims hence hold for $j_0 = 0$ here.

For $\ell > 3$ we can use $p > p_\ell > 3\ell^2$ to apply [LOX20, Theorem 2(2)], which directly yields claim (b) and further shows that – with $\left(\frac{\ell}{3}\right)$ denoting the Legendre symbol – there are exactly

$$(\ell + 1) - 3 \cdot \frac{1}{3} \left(\ell - \left(\frac{\ell}{3} \right) \right) = 1 + \left(\frac{\ell}{3} \right) \leq 2$$

non-equivalent ℓ -isogenies $0 \rightarrow 0$, thus also finishing the proof of claim (a).

Finally, for $j_0 = 1728 \neq j_1$ we have already shown that there are at most two non-equivalent ℓ -isogenies $1728 \rightarrow j_1$ and that 1728 is not ℓ -isogenous to 0. Thus the number of non-equivalent ℓ -isogenies $1728 \rightarrow j_1$ is exactly twice the number of non-equivalent ℓ -isogenies $j_1 \rightarrow 1728$, and claim (c) follows. $\square$

ℓ	2	3	5	7	11	13
p_ℓ	13	53	379	1217	5101	8387
j_0	5	$6a + 28$	$117a + 322$	$379a + 173$	$977a + 4220$	$326a + 4482$
j_1	5	$47a + 28$	$262a + 322$	$838a + 173$	$4124a + 4220$	$8061a + 4482$

Table 1: Maximal primes p_ℓ for which a pair (j_0, j_1) of non-zero j -invariants with at least three ℓ -isogenies between them exists (a is a square root of 349 modulo p_ℓ).

Remark 4. For any prime $\ell \in \mathbb{N}$ one can find a prime p_ℓ as in Theorem 9, and one has the bound $p_\ell < 4\ell^4$. Indeed, due to [LOX20, Theorem 2] it suffices to consider the situation where we have at least three non-equivalent ℓ -isogenies $j_0 \rightarrow j_1$ for $j_0 \notin \{0, 1728\}$ or $(j_0, j_1) = (0, 1728)$. In view of [BCNE⁺19, Theorem 4.10] we can then construct two non-commuting endomorphisms of degree ℓ^2 on a curve E with $j(E) = j_0$ by composing two non-equivalent ℓ -isogenies $j_0 \rightarrow j_1$ with a suitable ℓ -isogeny $j_1 \rightarrow j_0$. Hence we obtain two embeddings of quadratic orders into the endomorphism ring of E with distinct images, and these embeddings can be extended to optimal embeddings of (possibly larger) quadratic orders that still have distinct images as the endomorphisms do not commute. Thus Kaneko’s bound [Kan89, Theorem 2’] yields $4p_\ell \leq (-4\ell^2)^2$, i.e. $p_\ell < 4\ell^4$. In particular, by the discussion in Subsection 2.1 this proves that 0 and 1728 cannot be ℓ -isogenous for $p > 4\ell^4$.

Returning to our relation, let us suppose that we have two j -invariants $j_0, j_1 \in \overline{\mathbb{F}_p}$. As before we consider the polynomials

$$\Phi_\ell^c(X, j_0) \quad \text{and} \quad \Theta_\ell^c(X, j_1) = \Phi_\ell(\ell^s / X, j_1) \cdot X^{\ell+1} / \ell^s,$$

which have the same set of common solutions as the system (MT) due to $p \neq \ell$; these common solutions are, moreover, precisely the roots of the polynomial

$$\Gamma_\ell(j_0, j_1) := \gcd(\Phi_\ell^c(X, j_0), \Theta_\ell^c(X, j_1)) \in \mathbb{F}_p(j_0, j_1)[X].$$

The previous results now show that the degree of $\Gamma_\ell(j_0, j_1)$ is low in most cases:

Corollary 10. Let $\ell \in \{2, 3, 5, 7, 13\}$, let $j_0, j_1 \in \overline{\mathbb{F}_p}$ for a prime $p > \ell$, and consider as above the gcd-polynomial $\Gamma_\ell(j_0, j_1) \in \overline{\mathbb{F}_p}[X]$. Then

$$\deg \Gamma_\ell(j_0, j_1) = \min\{v_\ell(j_0, j_1), v_\ell(j_1, j_0)\}$$

where $v_\ell(j, j')$ denotes the number of non-equivalent ℓ -isogenies $j \rightarrow j'$. In particular:

- (a) $\deg \Gamma_\ell(j_0, j_1) \geq 1$ if and only if j_0 and j_1 are ℓ -isogenous.
- (b) If j_0 is ordinary or $p > p_\ell$ (e.g. if $p \geq 4\ell^3$), then $\deg \Gamma_\ell(j_0, j_1) \leq 2$.

Proof. The main claim is a direct consequence of the **Multiplicity Theorem** and the multiplicity analysis that was performed in its proof, noting that the minimum is necessary to account for higher root multiplicities at j -invariants 0 and 1728 (see also Lemma 7); hence claim (a) follows immediately. Moreover, Proposition 8 and Theorem 9 imply claim (b). $\square$

Remark 5. Since we can always factor an ℓ^2 -isogeny into two ℓ -isogenies (cf. [Sil09, Corollary III.4.11]), the j -invariants j_0 for which there is some j -invariant j_1 with $\deg \Gamma_\ell(j_0, j_1) \geq 2$ correspond precisely to the j -invariants of ℓ^2 -small curves as defined in [LB20], where the authors also prove that these j -invariants form a vanishingly small, but generally non-empty subset of $\overline{\mathbb{F}_p}$ for large p .

For $\ell \in \{2, 3, 5, 7, 13\}$ we can bound the size of this set more precisely: By factoring the J_1 -resultant

$$\text{res}_{J_1}(\Phi_\ell(J_0, J_1), \frac{\partial}{\partial J_1} \Phi_\ell(J_0, J_1)) \in \mathbb{Z}[J_0],$$

we see that the J_0 -degree sum of all its distinct irreducible factors is at most $\ell^2 + 1$ (cf. the script `additional_computations.sage`). Hence Proposition 1(a) shows that there are at most $\ell^2 + 1$ invariants $j_0 \in \overline{\mathbb{F}_p}$ that can belong to an ℓ^2 -small curve for $p > \ell$.

Computing roots. Corollary 10 suggests the following efficient strategy to find a root f of the system (MT) for two ℓ -isogenous j -invariants $j_0, j_1 \in \overline{\mathbb{F}_p}$: If the j -invariants do not lie in $\mathbb{F}_{p^2}$, and are hence necessarily ordinary, or if we have $p \geq 4\ell^3$ (which is guaranteed for cryptographically large primes), we can simply compute the gcd-polynomial $\Gamma_\ell(j_0, j_1)$ and obtain a root either by directly reading it off (in the degree 1 case) or by using the quadratic formula (in the degree 2 case, which only occurs rarely by Remark 5). Otherwise we will see in the next section (specifically Theorem 11) that $\Gamma_\ell(j_0, j_1) \in \mathbb{F}_{p^2}[X]$ splits into linear factors over $\mathbb{F}_{p^2}$ – this allows us to factor the polynomial over $\mathbb{F}_{p^2}$, e.g. using Berlekamp factorization. Note, however, that we are only interested in finding one root; thus we may, starting with $\Gamma_\ell(j_0, j_1)$, iteratively compute a partial factorization and only keep the factor of smallest degree at each step.

In view of the above results, we conclude that our new relation is practically equivalent to the relation $\mathcal{R}_{\ell^k\text{-MODPOLY}}$, i.e. a user with knowledge of an ℓ -isogeny j -invariant chain of length k can efficiently compute the additional roots $(f_i)_{i \in [k]}$ needed to prove their knowledge with respect to the relation $\mathcal{R}_{\ell^k\text{-MODROOT}}$.

3.3 Isogeny Reconstruction and Splitting Behavior

Where do the roots lie? After we have now given an essentially optimal bound on the number of roots of the system (MT) in large characteristic, we next investigate where these roots lie. Due to Remark 5 and the **Multiplicity Theorem**, this question is easily answered in the overwhelming majority of cases: For a j -invariant j_0 and a root f of $\Phi_\ell^c(X, j_0)$ such that there is only one non-equivalent ℓ -isogeny to $j_1 = \mathcal{J}_\ell(\ell^s/f)$, f is the unique root of the system given by the two polynomial equations $\Phi_\ell^c(X, j_0) = 0$ and $\Theta_\ell^c(\ell^s/X, j_1) = 0$, and hence lies in the field extension $\mathbb{F}_p(j_0, j_1)$. In fact, as $\mathcal{J}_\ell(f) = j_0$ and $\mathcal{J}_\ell(\ell^s/f) = j_1$ by Lemma 3, we have $\mathbb{F}_p(f) = \mathbb{F}_p(j_0, j_1)$.

To analyze the splitting behavior in general, however, additional work is required: As the following example shows, the field extension generated by the two j -invariants is not guaranteed to contain a root of the system (MT).

Example 2. Let $\ell = 3$, $p = 61$ and $j_0 = 9 \in \mathbb{F}_{61}$. Then we have the factorizations

$$\Phi_3(9, j_1) = (j_1 - 9)^2 \cdot (j_1 - 41)^2 \text{ and } \Phi_3^c(X, 9) = (X^2 - 15X - 3) \cdot (X^2 - 10X + 1)$$

over $\mathbb{F}_{61}$. Further we have $\Gamma_3(9, 9) = X^2 - 15X - 3$ and $\Gamma_3(9, 41) = X^2 - 10X + 1$; these polynomials are irreducible over $\mathbb{F}_{61}$, i.e. their roots lie in $\mathbb{F}_{61^2} \setminus \mathbb{F}_{61}$.

As a remedy to this issue, the following result describes how we can reconstruct an ℓ -isogeny from a common root f of the system (MT) in most situations:

Reconstruction Theorem. Let $\ell \in \{2, 3, 5, 7, 13\}$, let K be a field of characteristic $\text{char}(K) \notin \{2, 3, \ell\}$, let $j_0 \in K \setminus \{0, 1728\}$ be a j -invariant and let $f \in \overline{K}^\times$ be a root of $\Phi_\ell^c(X, j_0)$. Define the parameters

$$A = -3j_0(j_0 - 1728) \text{ and } B = -2j_0(j_0 - 1728)^2$$

and the elliptic curve

$$E: y^2 = x^3 + Ax + B.$$

Then $j(E) = j_0$, and we can associate to f a kernel polynomial $\phi_\ell(f)$ of degree $\left\lceil \frac{\ell-1}{2} \right\rceil$ that defines over $K(f)$ an ℓ -isogeny from E to a curve with j -invariant $\mathcal{J}_\ell(\ell^s/f)$. Moreover, f can be expressed as a K -rational function in the coefficients $1 = s_0, s_1, \dots, s_n$ of $\phi_\ell(f)$, i.e.

$$K(f) = K(s_1, \dots, s_n).$$

Proof. To limit confusion of variable names in this proof, we will write the canonical modular polynomial in the variables T and j (instead of the usual variables X and j). This proof will be highly computational – the observational claims used along the way can be verified via the SageMath script `kernel_polynomials.sage`.

The discriminant of the curve E is $2^{11} \cdot 3^5 \cdot j_0 \cdot (j_0 - 1728)$, which is non-zero by our assumptions. Hence E is an elliptic curve and one easily verifies that $j(E) = j_0$.

To work computationally, we will consider the coefficient ring $\mathbb{Z}[T, T^{-1}]$ of Laurent polynomials over $\mathbb{Z}$; since the root f is non-zero due to $\text{char}(K) \neq \ell$, we can then apply the ring homomorphism $\mathbb{Z}[T, T^{-1}] \rightarrow K(f)$ given by evaluating T at f . By Lemma 3 we can represent j_0 via the rational function

$$j_0(T) = \mathcal{J}_\ell(T) = \Phi_\ell^c(T, 0)/T;$$

now considering the coefficients of E as elements of $\mathbb{Z}[T, T^{-1}]$, and hence $E = E(T)$ as a curve over $\mathbb{Q}(T)$, we can apply the following deciding trick, which is based on the ideas in [CW05] and [Tsu13, §3-4]: The ℓ^{th} division polynomial $\psi_\ell \in \mathbb{Q}(T)[x]$ of $E(T)$ has coefficients in $\mathbb{Z}[T, T^{-1}]$ and admits in $\mathbb{Z}[T, T^{-1}][x]$ a monic factor $\phi_\ell \in \mathbb{Z}[T, T^{-1}][x]$ of degree $\left\lceil \frac{\ell-1}{2} \right\rceil =: n$. Evaluating at f hence yields the polynomial $\phi_\ell(f) \in K(f)[x]$ of degree n , and we want to show that this is a kernel polynomial.

For $\ell = 2$ we do this directly: Writing $\phi_2 = x - \xi$ and plugging the root ξ into the x -coordinate of $E(T)$ yields $y^2 = 0$, so $(\xi(f), 0)$ is a 2-torsion point of $E(f)$ as desired.

To prove that $\phi_\ell(f)$ is a kernel polynomial for $\ell \geq 3$, we want to apply the *Kernel polynomial criterion* given in [Tsu13, §3.3], noting that $a = 2$ is a *semi-primitive root modulo ℓ* as stated on [Tsu13, p. 34]. Hence we first have to compute the action $[2]^*(x)$ of the $[2]$ -endomorphism of E on the x -coordinate. By the multiplication-by-2 formula given in [Was08, Theorem 3.6] we have

$$[2]^*(x) = \frac{x^4 - 2A(T)x^2 - 8B(T)x + A(T)^2}{4x^3 + 4A(T)x + 4B(T)} =: \frac{t_1(T)}{t_2(T)}$$

where $\text{res}_x(t_1(T), t_2(T)) = 2^{22} \cdot 3^{10} \cdot j_0(T)^2 \cdot (j_0(T) - 1728)^2$ is the square of the discriminant of $E(T)$. In view of [Was08, Lemma 3.5], Proposition 1 thus shows that the polynomials $t_1(f)$ and $t_2(f)$ are coprime in $K(f)[x]$ due to our assumptions that $\text{char}(K) \notin \{2, 3\}$ and $j_0 \notin \{0, 1728\}$.

Next we have to evaluate ϕ_ℓ (in x) at $[2]^*(x)$, which gives

$$\phi_\ell\left(\frac{t_1(T)}{t_2(T)}\right) = \frac{1}{t_2(T)^n} \cdot \left[\phi_\ell\left(\frac{t_1(T)}{t_2(T)}\right) \cdot t_2(T)^n\right] =: \frac{1}{t_2(T)^n} \cdot h_\ell(T)$$

where $h_\ell(f)$ is coprime to $t_2(f)^n$ in $K(f)[x]$ since $t_1(f)$ and $t_2(f)$ are coprime. Now we define the monic polynomial

$$\tau_2(\phi_\ell(f)) := \gcd(\psi_\ell(f), h_\ell(f));$$

as $\psi_\ell(f)$ is the ℓ^{th} division polynomial of $E(f)$, we see with [Tsu13, Corollary 3.3.2] that $\phi_\ell(f)$ is a kernel polynomial of an ℓ -isogeny if and only if $\tau_2(\phi_\ell(f)) = \phi_\ell(f)$.

However, by [Tsu13, Proposition 3.3.1] we also see that $\tau_2(\phi_\ell(f))$ is a monic polynomial of degree $n = \frac{\ell-1}{2}$, so it suffices to show that $\phi_\ell(f)$ divides $\tau_2(\phi_\ell(f))$. To this end, first note that $\phi_\ell(T)$ divides $\psi_\ell(T)$ over $\mathbb{Z}[T, T^{-1}]$ by construction, which allows us to deduce that $\phi_\ell(f)$ divides $\psi_\ell(f)$.

Furthermore we can check computationally that $\phi_\ell(T)$ divides $h_\ell(T)$ over $\mathbb{Z}[T, T^{-1}]$, and hence $\phi_\ell(f)$ also divides $h_\ell(f)$. Therefore the definition of $\tau_2(\phi_\ell(f))$ forces it to be divisible by $\phi_\ell(f)$, and we conclude that $\phi_\ell(f)$ is a kernel polynomial of an ℓ -isogeny defined on $E(f)$.

Penultimately, we want to show that the isogeny defined by $\phi_\ell(f)$ maps to a curve of j -invariant $\mathcal{J}_\ell(\ell^s/f)$. This can be deduced directly from *V  lu's formulas* – more precisely, in [Koh96,  2.4] Kohel describes the target curve in terms of A, B and the coefficients of $\phi_\ell(f)$. Applying these formulas, we see that the target curve is in short Weierstrass form with discriminant $\Delta = 2^{11} \cdot 3^5 \cdot f^{\ell-1} \cdot A \cdot B \neq 0$ and j -invariant $\mathcal{J}_\ell(\ell^s/f)$.

Finally, we refer to Appendix A.4 for the (ℓ -dependent) expressions of f as a K -rational function in the coefficients of $\phi_\ell(f)$; here we only note that, as a kernel polynomial, $\phi_\ell(f)$ will never have any double roots, so $\text{disc}(\phi_\ell(f)) \neq 0$ by Proposition 1(b) (in view of Remark 1). $\square$

Remark 6. Note that we can also reconstruct ℓ -isogenies from a root $f \in \overline{K}^\times$ of $\Phi_\ell^c(X, j_0)$ if we have $j_1 = \mathcal{J}_\ell(\ell^s/f) \notin \{0, 1728\}$. Indeed, in this case we first compute the dual isogeny (up to equivalence) by applying the above techniques to ℓ^s/f , and then take its dual and precompose with the different automorphisms at j_0 to obtain the non-equivalent ℓ -isogenies $j_0 \rightarrow j_1$ corresponding to f .

We now use the *Reconstruction Theorem* to analyze the splitting behavior of $\Phi_\ell^c(X, j_0)$, i.e. where its roots lie, for a supersingular j -invariant j_0 .

Theorem 11. *Let $\ell \in \{2, 3, 5, 7, 13\}$, let $p \neq \ell$ be a prime and let $j_0 \in \mathbb{F}_{p^2}$ be a supersingular j -invariant. Then $\Phi_\ell^c(X, j_0)$ splits over $\mathbb{F}_{p^2}$.*

Proof. We first handle the case $j_0 \in \{0, 1728\}$: For $p \leq p_\ell$ we confirm the claim in the SageMath script `splitting_special.sage`, so we may assume $p > p_\ell$. By Theorem 9(b-c) and Corollary 10 we then see that $\Gamma_\ell(j_0, j_1) \in \mathbb{F}_{p^2}[X]$ has degree at most 1 for any supersingular $j_1 \neq j_0$, so the only roots of $\Phi_\ell^c(X, j_0)$ that may not lie in $\mathbb{F}_{p^2}$ are the roots of $\Gamma_\ell(j_0, j_0)$. However, by Corollary 10(b) this is a polynomial of degree at most 2 over $\mathbb{F}_p$.

Thus we may assume $j_0 \notin \{0, 1728\}$ and, in particular, $p \geq 5$. Now let $f \in \overline{\mathbb{F}_p}$ be a root of $\Phi_\ell^c(X, j_0)$. With the *Reconstruction Theorem* we can then associate to f the kernel polynomial $\phi_\ell(f)$ of an ℓ -isogeny defined on a curve E over $\mathbb{F}_{p^2}$.

Moreover, the kernel of this isogeny is invariant under the action of the p^2 -Frobenius isomorphism of $\overline{\mathbb{F}_p}$ on E . Indeed, this action is given by evaluation of the p^2 -Frobenius endomorphism π of E ; due to [AAM19,  4] we further see that π has trace $\pm 2p$ since $j_0 \notin \{0, 1728\}$, so it acts on E via scalar multiplication by $\pm p$ (cf. [AAM19,  5]) and we conclude that any subgroup of $E(\overline{\mathbb{F}_p})$ is invariant under the action of π .

Therefore the coefficients of the kernel polynomial $\phi_\ell(f)$ lie in $\mathbb{F}_{p^2}$ as well, and with the second part of the *Reconstruction Theorem* we deduce $f \in \mathbb{F}_{p^2}$ as desired. $\square$

We give an application, which can alternatively be proven by showing that any Legendre parameter of a supersingular j -invariant lies in $\mathbb{F}_{p^2}$ [AT02, Proposition 2.2].

Corollary 12. *Let p be a prime and let $j_0 \in \mathbb{F}_{p^2}$ be a supersingular j -invariant. Then $j_0 - 1728$ is a square in $\mathbb{F}_{p^2}$.*

Proof. As all elements of $\mathbb{F}_p$ are squares in $\mathbb{F}_{p^2}$, we may assume $j_0 \notin \{0, 1728\}$ and, in particular, $p \geq 5$. Due to Proposition 1(a) we can thus compute the discriminant of $\Phi_2^c(X, j_0)$ by first computing the discriminant of $\Phi_2^c(X, J_0)$ over $\mathbb{Z}[J_0]$ and then reducing modulo p and evaluating J_0 at j_0 , which yields

$$\text{disc}(\Phi_2^c(X, j_0)) = 2^2 \cdot j_0^2 \cdot (j_0 - 1728).$$

Now $\Phi_2^c(X, j_0)$ does not have multiple roots by Lemma 6 and splits into three linear factors over $\mathbb{F}_{p^2}$ by Theorem 11, so [Gow90, Theorem 1.8] shows that $\text{disc}(\Phi_2^c(X, j_0))$ must be a square in $\mathbb{F}_{p^2}$, and the claim follows. $\square$

4 Proving Isogeny Knowledge via R1CS

In the previous section we have laid all the theoretical foundations for describing the R1CS that will enable us to build an efficient proof of knowledge for an ℓ -isogeny walk with k steps. Before we describe our approach based on canonical modular polynomials, we briefly revisit the strategy pursued in [CLL23] for the prime $\ell = 2$.

4.1 Revisiting the Approach in [CLL23]

The authors of [CLL23] use the classical modular polynomial $\Phi_2(X, Y)$ to construct an R1CS to prove knowledge of a 2^k -isogeny with respect to the relation $\mathcal{R}_{\ell^k\text{-MODPOLY}}$ (cf. Equation (1)). They do this by finding an efficient arithmetization to prove that $\Phi_2(j_i, j_{i+1}) = 0$ for a chain of $k + 1$ successive j -invariants. Here $j_0 = j(E_0)$ and $j_k = j(E_k)$ are part of the statement, and j_i for $0 < i < k$ are part of the R1CS witness. We can recover the original isogeny by searching at each step for 2-isogenous elliptic curves E_i, E_{i+1} where $j(E_i) = j_i$ and $j(E_{i+1}) = j_{i+1}$. On the other hand, such a chain of j -invariants can be found for any 2^k -isogeny by iteratively computing 2-isogenies using kernel points. This means that the problem of finding such a chain of j -invariants is equivalent to finding an explicit isogeny $E_0 \rightarrow E_k$ of degree 2^k .

In order to arithmetize, the authors express each step of the isogeny walk as an R1CS gadget, which is then employed for each link in the chain. Two tricks are used to optimize:

- The values j_i, j_i^2 and j_i^3 are computed for all $i \in \{0, \dots, k\}$, as well as $j_i j_{i+1}$ for each $i < k$. The condition that $\Phi_2(j_i, j_{i+1}) = 0$ can then be expressed as a single R1CS constraint.
- To express the gadget over $\mathbb{F}_p$ as well as over $\mathbb{F}_{p^2}$, the authors use arithmetizations for products and squares that are more efficient than the naive approach of computing each cross term individually. In particular, the product $(x_1 + x_2\alpha)(y_1 + y_2\alpha)$, with $x_1, x_2, y_1, y_2 \in \mathbb{F}_p$ and $\alpha^2 = d$ some quadratic non-residue in $\mathbb{F}_p$, can be expressed in three products over $\mathbb{F}_p$. Squarings can be expressed in two products.

Our goal is now twofold: First, to further optimize the arithmetization for $\ell = 2$. Second, to construct efficient R1CS for more primes $\ell > 2$, more specifically for the primes $\ell \in \{2, 3, 5, 7, 13\}$, for which we have developed a good understanding of canonical modular polynomials in the previous section.

By Lemma 3, for $\ell \in \{2, 3, 5, 7, 13\}$ the ℓ^{th} canonical modular polynomial has the form

$$\Phi_\ell^c(X, j) = X^{\ell+1} + \sum_{i=1}^{\ell} c_i X^i + \ell^s - X \cdot j.$$

In what follows we will write $c_0 = \ell^s$ and $c_{\ell+1} = 1$. In view of the **Multiplicity Theorem** the proof of knowledge with respect to relation $\mathcal{R}_{\ell^k\text{-MODROOT}}$ can be encoded step-wise via the system of equations (MT).

Multiplying the equation $\Phi_\ell^c(\ell^s/X, j_1) = 0$ by $X^{\ell+1}/\ell^s$ to obtain $\Theta_\ell^c(X, j_1)$ as before, we obtain the equivalent system (where $c'_i = c_{\ell+1-i} \cdot \ell^{s(\ell-i)}$):

$$\sum_{i=0}^{\ell+1} c_i X^i - j_0 \cdot X = 0 \quad \wedge \quad \sum_{i=0}^{\ell+1} c'_i X^i - j_1 \cdot X^\ell = 0. \quad (4)$$

We will reformulate these equations as an R1CS in the upcoming subsection. In our applications we consider supersingular j -invariants, which are known to be contained in $\mathbb{F}_{p^2}$. It is crucial for the effectivity of our method that in this situation the roots of the above equations still lie in the quadratic extension $\mathbb{F}_{p^2}$ of $\mathbb{F}_p$, rather than in a larger extension, as proven in Theorem 11.

4.2 Reformulation as an R1CS

Advantages of Canonical Modular Polynomials. Compared to the classical modular polynomials, the canonical modular polynomials have a structure that lends itself better to arithmetization. Concretely, as we will see, we can exploit their structure for the R1CS in three main ways:

- First, the total degree of the polynomials is lower, going from a single polynomial of total degree 2ℓ to two polynomials of degree $\ell + 1$. This lowers the multiplicative complexity, which enables us to use fewer constraints.

- Second, whereas Φ_ℓ is very dense, Φ_ℓ^c and Θ_ℓ^c are both polynomials in just X in addition to a single term containing j . Hence there are fewer monomials to produce in the R1CS and the computed terms can more often be reused.
- Lastly, the structure described in Lemma 7 allows us to factor part of this polynomial as a square, improving arithmetization over $\mathbb{F}_p$.

More efficient R1CS over $\mathbb{F}_{p^2}$. We can compute the powers $1, X, X^2, \dots, X^\ell$ together with the j -invariants j and j' , and rewrite the equations as

$$X \cdot \left(\sum_{i=0}^{\ell} c_{i+1} X^i - j \right) + c_0 = 0, \quad (5)$$

$$X^\ell \cdot \left(\sum_{i=0}^1 c'_{\ell+i} X^i - j' \right) + \sum_{i=0}^{\ell-1} c'_i X^i = 0. \quad (6)$$

To reduce the number of non-zero entries, we employ a change of variables and have the prover supply $y = j - c_1$ instead of j and $y' = j' - c_1$ instead of j' . This eliminates the term X from the first equation and the term X^ℓ from the second equation, since $c_1 = c'_\ell$. Clearly knowledge of a chain of j -invariants is equivalent to knowledge of a chain of y 's.

These equations are expressed as an R1CS as follows. The assignment vector z has the form $z = (1 \ X \ X^2 \ \dots \ X^\ell \ y \ y')^T$, and the corresponding constraint matrices are given by

$$A = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & & & & & & \\ 0 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 1 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & & & & & & \\ 0 & c_2 & c_3 & \cdots & c_{\ell+1} & -1 & 0 \\ 0 & c'_{\ell+1} & 0 & \cdots & 0 & 0 & -1 \end{pmatrix},$$

and

$$C = \begin{pmatrix} 0 & 0 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & & & & & & & \\ 0 & 0 & 0 & 0 & \cdots & 1 & 0 & 0 \\ -c_0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ -c'_0 & -c'_1 & -c'_2 & -c'_3 & \cdots & -c'_{\ell-1} & 0 & 0 \end{pmatrix}.$$

For an isogeny path of length k , each new step introduces a new value y_{i+1} together with the ℓ powers of the current solution X_i . Moreover, we need $\ell - 1$ more constraints for checking the powers of the new variable X_i , together with two more constraints which encode the two equations (4). There are $\ell + 1$, $2\ell + 2$ and 2ℓ non-zero entries in A , B and C , respectively. This means that we can arithmetize an ℓ^k -isogeny for $\ell \in \{2, 3, 5, 7, 13\}$ in $(\ell + 1)k + 1$ variables, $(\ell + 1)k$ constraints and $(5\ell + 3)k$ non-zero entries in the R1CS.

In fact, we can do better for $\ell \in \{7, 13\}$, in the following way. We let $t = \frac{\ell+1}{2}$ and rewrite the two equations as

$$X^t \cdot \left(\sum_{i=0}^t c_{i+t} X^i \right) + \sum_{i=0, i \neq 1}^{t-1} c_i X^i - yX = 0, \quad (7)$$

$$X^t \cdot \left(\sum_{i=0, i \neq t-1}^t c'_{i+t} X^i - y' X^{t-1} \right) + \sum_{i=0}^{t-1} c'_i X^i = 0. \quad (8)$$

This way we need to compute the variables $X^2, \dots, X^t, yX, y'X^{t-1}$, and we have $t + 1 + 2$ constraints ($t + 1$ consistency checks, and the two above equations). We have t variables for the powers of X and two for yX and $y'X^{t-1}$, as well as one per j -invariant, through y . This gives a total of $(t + 3)k + 1$ variables. The $t + 1$ consistency checks can be computed in $3t + 3$, while the two other equations require $4t + 4$ non-zero entries in the constraint matrices. In total, this means $(t + 3)k + 1$ variables, $(t + 3)k$ constraints and $(7t + 7)k$ non-zero entries in the R1CS to arithmetize a walk of length k for $\ell \in \{3, 5, 7, 13\}$.

4.3 Lifting to $\mathbb{F}_p \times \mathbb{F}_p$

So far, we have described efficient arithmetizations for proving isogenies that work over $\mathbb{F}_{p^2}$. While it is perfectly valid to use an R1CS over this field, this is not supported by all proof systems. Fortunately, we can lift any arithmetization over $\mathbb{F}_{p^2}$ to $\mathbb{F}_p \times \mathbb{F}_p$.

We represent elements in $\mathbb{F}_{p^2}$ as $x + y\alpha$ with $x, y \in \mathbb{F}_p$ and $\alpha^2 = d$ for some quadratic non-residue $d \in \mathbb{F}_p$. Linear operations over $\mathbb{F}_{p^2}$ translate directly to linear operations over $\mathbb{F}_p$, whereas a multiplication $(x_1 + x_2\alpha)(y_1 + y_2\alpha) = z_1 + z_2\alpha$ would naively induce four multiplications for all cross terms x_1y_1 , $x_1y_2\alpha$, $x_2y_1\alpha$ and x_2y_2d . As noted in [CLL23], there exist well-known techniques to do this more efficiently. Multiplication can be performed using one auxiliary variable and three constraints by applying Karatsuba's algorithm:

$$\begin{aligned} u &= x_2y_2, \\ z_1 - du &= x_1y_1, \\ z_1 + z_2 + (1 - d)u &= (x_1 + x_2)(y_1 + y_2). \end{aligned}$$

This immediately implies an upper bound for the cost of lifting: We can substitute this relation for every row of the original R1CS to obtain a new system with $m' = 3m$. The number of variables becomes $n' = 2n + m$, since variables now have two components and we add one intermediate variable for every constraint. The number of non-zero entries for this new system is $nnz' \leq 4nnz$ due to the doubling of the number of variables, and because all variables are used at most twice in the above system of equations.

Squaring a variable can be performed more efficiently. For

$$z_1 + z_2\alpha = (x_1 + x_2\alpha)^2,$$

the following system of equations suffices:

$$\begin{aligned} z_2 &= 2x_1x_2, \\ z_1 + (d + 1)/2z_2 &= (x_1 + x_2)(x_1 + dx_2). \end{aligned}$$

This uses just two constraints and nine non-zero entries, requiring no auxiliary variables.

Optimized embedding into $\mathbb{F}_p$. The above gadgets for multiplication and squaring can be used to reduce the number of constraints needed in the R1CS. However, we can actually achieve a further improvement by changing how we represent elements from $\mathbb{F}_{p^2}$ in $\mathbb{F}_p \times \mathbb{F}_p$. The basis change $(x_1, x_2) \rightarrow (x_1, x_1 + x_2) = (x_1, x_s)$ saves one non-zero entry in the last constraint for the general product. This might seem minor, but since both coefficients of x can themselves be a linear combination of many variables, this may lead to significant savings.

For a square, instead of avoiding a non-zero entry, we need to add an entry to express the output $z_2 = z_s - z_1$. Fortunately, although clearly this gives a tradeoff between the two bases, the embedding can be chosen separately per variable. In our case, this sum basis turns out to be at least as efficient as the regular basis almost everywhere.

Exploiting Φ_ℓ^c 's structure. The above gadgets and the sum basis help us to express the canonical modular polynomials efficiently in an R1CS over $\mathbb{F}_p \times \mathbb{F}_p$. Furthermore, for some ℓ , the canonical modular polynomial can be rewritten using the structure described in Lemma 7 such that it utilizes more squares. This in turn minimizes the number of necessary constraints. For $\ell = 2$ over $\mathbb{F}_p \times \mathbb{F}_p$, we can write the entire system using two squares and one product. First, define $y = c_1 - c_2^2 \cdot (4c_3)^{-1} - j$ (and y' accordingly). Then

$$\begin{aligned} c_3 \left(c_2 \cdot (2c_3)^{-1} + X \right)^2 + y + c_0X^{-1} &= 0, \\ c_3\ell^{3s} \left(c_2\ell^s \cdot (2c_3)^{-1} + X^{-1} \right)^2 + \ell^s y' + c_0X &= 0, \end{aligned}$$

captures both polynomials in two squaring relations. We only need to compute the inverse X^{-1} with a single additional multiplication. Over $\mathbb{F}_p \times \mathbb{F}_p$, the resulting system has $7k$ constraints and $7k + 2$ variables. In a similar fashion, for $\ell = 7$, we can use the fact that Equation (7) can be written as a square plus the term $\tilde{y} = 1728 - j$. For $\ell = 13$, we can express 9 coefficients correctly using the square of a degree 7 polynomial, after which we only need to correct for the lowest degree terms.

Computing powers of X . To compute even powers, we can square directly. It is, however, also possible to use a squaring for odd powers, through the relation

$$b \left(a \cdot (2b)^{-1} X^i + X^{i+1} \right)^2 = a^2 \cdot (4b)^{-1} X^{2i} + a X^{2i+1} + b X^{2i+2}. \quad (9)$$

We hence obtain a linear combination of three powers, where we can freely choose $a \in \mathbb{F}_p$ and $b \in \mathbb{F}_p^\times$. By subtracting away $a^2 \cdot (4b)^{-1} X^{2i}$ and $b X^{2i+2}$, we obtain a constraint for the odd power X^{2i+1} . Of course, this method cannot be used for the products with y or the highest power of X , i.e. X^ℓ or X^t , since here we cannot compensate for the even powers appearing on the right-hand side.

Change of variables. One disadvantage of the above method is that it increases the number of non-zero entries in the constraint matrices. To remedy this, we note that some powers of X are only used in linear combinations with other powers, i.e. in a polynomial where all three powers are already present. As such, we make a change of variables and store the right-hand side of Equation (9) directly instead of X^{2i+1} . We can then choose a and b appropriately such that they agree with the coefficients of one of the polynomials. For example, for $\ell = 5$ we define $Z = c_5(c_4 \cdot (2c_5)^{-1} X + X^2)^2$ and rewrite Equation (5) as

$$X \cdot \left[c_2 X + \left(c_3 - c_4^2 \cdot (4c_5)^{-1} \right) X^2 + Z + X^5 - y \right] + c_0 = 0. \quad (10)$$

The advantage is twofold: We do not need to subtract powers from Equation (9) when computing X^3 , and we do not have to add an X^4 term to Equation (10). We should only use this substitution for powers that are not necessary to compute higher powers: The above would be inefficient if we also required the value of X^3 directly to express $(X^3)^2 = X^6$. Concretely, we use this trick to replace X^3 for $\ell \in \{5, 7\}$ and X^5 for $\ell = 13$.

More generally, the number of non-zero entries can often be minimized through a change of variables. For example, since jX and $j'X^{t-1}$ are both only used once, we can instead already add the terms of the linear combinations in which they will be used later. This is advantageous, since a linear combination uses fewer nnz in the outcome of a square than in its input, and is still cheaper in the outcome of a multiplication. This way, the intermediate variables that are unavoidable can be used as efficiently as possible.

4.4 Optimized Backtracking Prevention

In [CLL23, Appendix A], the authors briefly discuss a method of enforcing that an isogeny is cyclic, which in our setting is equivalent to the condition that a walk is non-backtracking (cf. [CFL⁺19, Corollary 4.5]).

To do this, it is checked that for all $i \in \{1, \dots, k-1\}$, it holds that $j_{i-1} \neq j_{i+1}$. The authors arithmetize this by testing that $j_{i-1} - j_{i+1} \neq 0$. To this end, the prover is required to supply a b such that

$$b \cdot \prod_{i=1}^{k-1} (j_{i-1} - j_{i+1}) = b(j_0 - j_2)(j_1 - j_3)(j_2 - j_4) \cdots = 1.$$

If for any i , $j_{i-1} = j_{i+1}$, then the polynomial will be 0 for any b , and otherwise the prover can solve for b by taking an inverse. This method uses $k-1$ constraints, $k-1$ variables and $4k-4$ non-zero entries over $\mathbb{F}_{p^2}$. When this method is lifted to $\mathbb{F}_p \times \mathbb{F}_p$, however, it uses $3(k-1)$ constraints and variables, and $18k-19$ non-zero entries. If we were to apply it to our system for $\ell = 2$, this would incur a 43% overhead on m and n , and a 44% overhead on nnz.

To remedy this, note that it is not actually necessary to compute a product over $\mathbb{F}_{p^2}$ to test that every factor is non-zero. Instead, we can use the constraint

$$p(a, b) = b \prod_{i=1}^{k-1} (\Re(j_{i-1}) - \Re(j_{i+1}) + a(\Im(j_{i-1}) - \Im(j_{i+1}))) = 1.$$

If $j_{i-1} = j_{i+1}$ for any i , $p(a, b) = 0$. On the other hand, if for all i , $j_{i-1} \neq j_{i+1}$, then $p(a, b)$ is a non-zero polynomial. Fixing $b \neq 0$ and applying the Schwartz-Zippel lemma, there are at least $p - k + 1$ values of a for which $p(a, b) \neq 0$, and for each such a there then is a suitable b to satisfy the constraint.

We can either let the prover choose $a, b \in \mathbb{F}_p$ for perfect soundness and completeness, or let a be a uniformly random constant. Now the prover just has to supply b , saving one constraint and variable per step. This incurs no soundness loss, since no value of a helps a cheating prover. On the other hand, there is at most a $\frac{k-1}{p} \leq \text{negl}(\lambda)$ completeness loss. Finally, we can set $a = 0$ and save two nnz per step, but now the completeness loss depends on the distribution of the path’s j -invariants.

5 Evaluation

We provide `constraints.sage`, which expresses and verifies all arithmetizations and automatically counts the number of constraints, variables and nnz. These results can be found in Table 2. We achieve significant improvements everywhere, ranging between 25% – 45% for the number of constraints and 27% – 48% for nnz, making our arithmetizations suitable for ℓ -power isogenies for each prime $\ell \in \{2, 3, 5, 7, 13\}$, as well as mixed power isogenies. To additionally compare results for distinct ℓ , we normalize by considering a security level λ , such that $\ell^k > 2^\lambda$. By increasing ℓ we can decrease k , reducing the number of constraints necessary for the relation $\mathcal{R}_{\text{R1CS}}$. These results can be found in Table 3. We see that moving to higher ℓ further improves efficiency for walks of similar degree.

In Table 4 we provide concrete prover and verifier times as well as proof sizes for Aurora [BCR⁺19] and Liger [AHIV17]. We use the parameter sets p434 and p441+ from [CLL23].⁴ These results are projected based on the number of constraints m , which appears to be a common and reliable basis for comparison.⁵

ℓ	Field	m		n		nnz	
		[CLL23]	Ours	[CLL23]	Ours	[CLL23]	Ours
2	$\mathbb{F}_{p^2}$	$4k + 2$	$3k$	$4k + 3$	$3k + 1$	$21k + 6$	$13k$
3			$4k$		$4k + 1$		$18k$
5			$6k$		$6k + 1$		$28k$
7			$7k$		$7k + 1$		$35k$
13			$10k$		$10k + 1$		$56k$
2	$\mathbb{F}_p$	$11k + 5$	$7k$	$11k + 7$	$7k + 2$	$79k + 23$	$41k$
3			$11k$		$11k + 2$		$65k$
5			$15k$		$15k + 2$		$97k$
7			$17k$		$17k + 2$		$123k$
13			$24k$		$24k + 2$		$194k$

Table 2: Our results compared to [CLL23]. We consider the number of constraints m , the number of variables n and the number of non-zero entries in the constraint matrices nnz.

ℓ	Field	m		n		nnz	
		[CLL23]	Ours	[CLL23]	Ours	[CLL23]	Ours
2	$\mathbb{F}_{p^2}$	$4\lambda + 2$	3λ	$4\lambda + 3$	$3\lambda + 1$	$21\lambda + 6$	13λ
3			2.524λ		$2.524\lambda + 1$		11.357λ
5			2.584λ		$2.584\lambda + 1$		12.059λ
7			2.493λ		$2.493\lambda + 1$		12.467λ
13			2.702λ		$2.702\lambda + 1$		15.133λ
2	$\mathbb{F}_p$	$11\lambda + 5$	7λ	$11\lambda + 7$	$7\lambda + 2$	$79\lambda + 23$	41λ
3			6.940λ		$6.940\lambda + 2$		41.010λ
5			6.460λ		$6.460\lambda + 2$		41.776λ
7			6.056λ		$6.056\lambda + 2$		43.813λ
13			6.486λ		$6.486\lambda + 2$		52.426λ

Table 3: Our results compared to those of [CLL23], normalized for security parameter λ .

⁴ Measurements were done on a desktop PC on a single thread and based on academic implementation libiop. Because of this, we expect optimizations and multithreading to yield significant improvements.

⁵ See e.g. [BCR⁺19], [GLS⁺23] and <https://github.com/scipr-lab/libiop> for independent benchmarks.

Field		Aurora		Ligero	
		[CLL23]	Ours	[CLL23]	Ours
$\mathbb{F}_{p^2}$	Prover time (ms)	934	669	587	420
	Verifier time (ms)	99	74	847	634
	Proof size (kB)	194	178	1849	1599
$\mathbb{F}_p$	Prover time (ms)	1216	727	427	255
	Verifier time (ms)	98	62	493	313
	Proof size (kB)	166	147	1733	1381

Table 4: An evaluation of our results compared to [CLL23] for walks with $\ell = 2$. The results from $\mathbb{F}_{p^2}$ correspond to the identification protocol with parameter set p434, with path length $k = 216$ and security level $\lambda = 128$, while $\mathbb{F}_p$ corresponds to p441+. The timings and proof sizes are from [CLL23] and subsequently projected from m .

Finally, our additional circuit to prevent backtracking with variable a , reduces m by one-third. The approach with constant a and negligible completeness loss reduces m by two-thirds. The full overview is found in Table 5.

Field	Approach	m	n	nnz
$\mathbb{F}_{p^2}$	[CLL23]	$k - 1$	$k - 1$	$4k - 4$
$\mathbb{F}_p$	[CLL23]	$3k - 3$	$3k - 3$	$18k - 19$
	Ours, variable a	$2k - 2$	$2k - 1$	$9k - 9$
	Ours, constant a	$k - 1$	$k - 1$	$6k - 6$
	Ours, $a = 0$	$k - 1$	$k - 1$	$4k - 4$

Table 5: The number of constraints m , variables n and number of non-zero entries nnz used to prevent backtracking using the existing method from [CLL23] and our methods. For variables, we do not include 1 and the j -invariants in the count.

6 Conclusion and Open Problems

In this paper we improved on the state-of-the-art of using general-purpose zero-knowledge proof systems for proving knowledge of an isogeny via R1CS. We were able to generalize the approach of Cong, Lai and Levin [CLL23] beyond $\ell = 2$ to prime numbers $\ell \in \{3, 5, 7, 13\}$ via the use of canonical modular polynomials. Moreover, we optimized the arithmetizations for the corresponding relation both over $\mathbb{F}_{p^2}$ and over $\mathbb{F}_p \times \mathbb{F}_p$.

In the course of our work we encountered interesting mathematical questions, some of which might hold in greater generality. For example, while Remark 4 argues that one can generalize Theorem 9 to any prime ℓ with the looser bound $p_\ell < 4\ell^4$, it is natural to wonder whether the tighter bound $p_\ell < 4\ell^3$ holds for any ℓ – in fact, between the first and the current version of this article, this bound has been achieved in [BS25, Corollary 16 & Proposition 30] in view of [LOX20, Theorem 2].

It might be even more interesting to study the canonical modular polynomials (or different, equivalent polynomials) for primes ℓ such that $\kappa > 1$. In that case we do not know to what extent the Multiplicity Theorem still holds true. More precisely, we expect one inequality to still hold, but the other to fail generally, as discussed in Remark 2.

Acknowledgements. The last author would like to thank Jonathan Love, Travis Morrison and Eli Orvis for interesting discussions on the supersingular isogeny graph, which led to Remark 4, at the Leuven Isogeny Days 5. The authors further want to thank all anonymous reviewers for their feedback.

References

- [AAM19] Gora Adj, Omran Ahmadi, and Alfred Menezes. On isogeny graphs of supersingular elliptic curves over finite fields. *Finite Fields and Their Applications*, 55:268–283, 2019.
- [ABCP23a] Shahla Atapoor, Karim Bagheri, Daniele Cozzo, and Robi Pedersen. Practical robust DKG protocols for CSIDH. In Mehdi Tibouchi and Xiaofeng Wang, editors, *ACNS 2023: 21st International Conference on Applied Cryptography and Network Security, Part II*, volume 13906 of *Lecture Notes in Computer Science*, pages 219–247, Kyoto, Japan, June 19–22, 2023. Springer, Cham, Switzerland.
- [ABCP23b] Shahla Atapoor, Karim Bagheri, Daniele Cozzo, and Robi Pedersen. VSS from distributed ZK proofs and applications. In Jian Guo and Ron Steinfeld, editors, *Advances in Cryptology – ASIACRYPT 2023, Part I*, volume 14438 of *Lecture Notes in Computer Science*, pages 405–440, Guangzhou, China, December 4–8, 2023. Springer, Singapore, Singapore.
- [AGRS24] Behzad Abdolmaleki, Noemi Glaeser, Sebastian Ramacher, and Daniel Slamanig. Circuit-succinct universally composable nizks with updatable crs. In *2024 IEEE 37th Computer Security Foundations Symposium (CSF)*, pages 527–542, Los Alamitos, CA, USA, jul 2024. IEEE Computer Society.
- [AHIV17] Scott Ames, Carmit Hazay, Yuval Ishai, and Muthuramakrishnan Venkatasubramanian. Liger: Lightweight sublinear arguments without a trusted setup. In Bhavani M. Thuraisingham, David Evans, Tal Malkin, and Dongyan Xu, editors, *ACM CCS 2017: 24th Conference on Computer and Communications Security*, pages 2087–2104, Dallas, TX, USA, October 31 – November 2, 2017. ACM Press.
- [AT02] Roland Auer and Jaap Top. Legendre elliptic curves over finite fields. *Journal of Number Theory*, 95(2):303–312, 2002.
- [BBBF18] Dan Boneh, Joseph Bonneau, Benedikt Bünz, and Ben Fisch. Verifiable delay functions. In Hovav Shacham and Alexandra Boldyreva, editors, *Advances in Cryptology – CRYPTO 2018, Part I*, volume 10991 of *Lecture Notes in Computer Science*, pages 757–788, Santa Barbara, CA, USA, August 19–23, 2018. Springer, Cham, Switzerland.
- [BCC⁺23] Andrea Basso, Giulio Codogni, Deirdre Connolly, Luca De Feo, Tako Boris Fouotsa, Guido Maria Lido, Travis Morrison, Lorenz Panny, Sikhar Patranabis, and Benjamin Wesolowski. Supersingular curves you can trust. In Carmit Hazay and Martijn Stam, editors, *Advances in Cryptology – EUROCRYPT 2023, Part II*, volume 14005 of *Lecture Notes in Computer Science*, pages 405–437, Lyon, France, April 23–27, 2023. Springer, Cham, Switzerland.
- [BCNE⁺19] Efrat Bank, Catalina Camacho-Navarro, Kirsten Eisenträger, Travis Morrison, and Jennifer Park. Cycles in the supersingular ℓ -isogeny graph and corresponding endomorphisms. In Jennifer S. Balakrishnan, Amanda Folsom, Matilde Lalín, and Michelle Manes, editors, *Research Directions in Number Theory. Women in Numbers IV*, volume 19 of *Association for Women in Mathematics Series*, pages 41–66. Springer Cham, 2019.
- [BCR⁺19] Eli Ben-Sasson, Alessandro Chiesa, Michael Riabzev, Nicholas Spooner, Madars Virza, and Nicholas P. Ward. Aurora: Transparent succinct arguments for R1CS. In Yuval Ishai and Vincent Rijmen, editors, *Advances in Cryptology – EUROCRYPT 2019, Part I*, volume 11476 of *Lecture Notes in Computer Science*, pages 103–128, Darmstadt, Germany, May 19–23, 2019. Springer, Cham, Switzerland.
- [BD21] Jeffrey Burdges and Luca De Feo. Delay encryption. In Anne Canteaut and François-Xavier Standaert, editors, *Advances in Cryptology – EUROCRYPT 2021, Part I*, volume 12696 of *Lecture Notes in Computer Science*, pages 302–326, Zagreb, Croatia, October 17–21, 2021. Springer, Cham, Switzerland.

- [BDFGP23] Ward Beullens, Luca De Feo, Steven D. Galbraith, and Christophe Petit. Proving knowledge of isogenies: a survey. *Designs, Codes and Cryptography*, 91(11):3425–3456, 2023.
- [BKV19] Ward Beullens, Thorsten Kleinjung, and Frederik Vercauteren. CSI-FiSh: Efficient isogeny based signatures through class group computations. In Steven D. Galbraith and Shiho Moriai, editors, *Advances in Cryptology – ASIACRYPT 2019, Part I*, volume 11921 of *Lecture Notes in Computer Science*, pages 227–247, Kobe, Japan, December 8–12, 2019. Springer, Cham, Switzerland.
- [BM25] Andrea Basso and Luciano Maino. POKÉ: A compact and efficient PKE from higher-dimensional isogenies. In Serge Fehr and Pierre-Alain Fouque, editors, *Advances in Cryptology – EUROCRYPT 2025, Part II*, volume 15602 of *Lecture Notes in Computer Science*, pages 94–123, Madrid, Spain, May 4–8, 2025. Springer, Cham, Switzerland.
- [Bos18] Siegfried Bosch. *Algebra. From the Viewpoint of Galois Theory*. Birkhäuser Advanced Texts Basler Lehrbücher. Birkhäuser Cham, 2018.
- [BPR06] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*, volume 10 of *Algorithms and Computation in Mathematics*. Springer Berlin, Heidelberg, 2nd edition, 2006.
- [BS23] Ward Beullens and Gregor Seiler. LaBRADOR: Compact proofs for R1CS from module-SIS. In Helena Handschuh and Anna Lysyanskaya, editors, *Advances in Cryptology – CRYPTO 2023, Part V*, volume 14085 of *Lecture Notes in Computer Science*, pages 518–548, Santa Barbara, CA, USA, August 20–24, 2023. Springer, Cham, Switzerland.
- [BS25] Sebastiano Boscardin and Sebastian A. Spindler. Exploring Kaneko’s bound: On multi-edges, loops and the diameter of the supersingular ℓ -isogeny graph. *Cryptology ePrint Archive*, Paper 2025/1361, 2025.
- [CFA⁺06] Henri Cohen, Gerhard Frey, Roberto M. Avanzi, Christophe Doche, Tanja Lange, Kim Nguyen, and Frederik Vercauteren. *Handbook of elliptic and hyperelliptic curve cryptography*. Discrete Math. Appl. (Boca Raton). Chapman & Hall/CRC, Boca Raton, FL, 2006.
- [CFL⁺19] Annamaria Costache, Brooke Feigon, Kristin Lauter, Maike Massierer, and Anna Puskás. Ramanujan graphs in cryptography. In Jennifer S. Balakrishnan, Amanda Folsom, Matilde Lalin, and Michelle Manes, editors, *Research Directions in Number Theory. Women in Numbers IV*, volume 19 of *Association for Women in Mathematics Series*, pages 1–40. Springer Cham, 2019.
- [CL24] Giulio Codogni and Guido Lido. Spectral theory of isogeny graphs. [arXiv:2308.13913v3 \[math.NT\]](https://arxiv.org/abs/2308.13913v3), July 2024.
- [CLG09] Denis X. Charles, Kristin E. Lauter, and Eyal Z. Goren. Cryptographic hash functions from expander graphs. *Journal of Cryptology*, 22:93–113, 2009.
- [CLL23] Kelong Cong, Yi-Fu Lai, and Shai Levin. Efficient isogeny proofs using generic techniques. In Mehdi Tibouchi and Xiaofeng Wang, editors, *ACNS 2023: 21st International Conference on Applied Cryptography and Network Security, Part II*, volume 13906 of *Lecture Notes in Computer Science*, pages 248–275, Kyoto, Japan, June 19–22, 2023. Springer, Cham, Switzerland.
- [CLM⁺18] Wouter Castryck, Tanja Lange, Chloe Martindale, Lorenz Panny, and Joost Renes. CSIDH: An efficient post-quantum commutative group action. In Thomas Peyrin and Steven Galbraith, editors, *Advances in Cryptology – ASIACRYPT 2018, Part III*, volume 11274 of *Lecture Notes in Computer Science*, pages 395–427, Brisbane, Queensland, Australia, December 2–6, 2018. Springer, Cham, Switzerland.
- [Cou06] Jean-Marc Couveignes. Hard homogeneous spaces. *Cryptology ePrint Archive*, Paper 2006/291, 2006.

- [Cox13] David A. Cox. *Primes of the Form $x^2 + ny^2$. Fermat, Class Field Theory, and Complex Multiplication*. Pure and Applied Mathematics: A Wiley Series of Texts, Monographs and Tracts. John Wiley & Sons, 2nd edition, 2013.
- [CSF⁺08] D. Cooper, S. Santesson, S. Farrell, S. Boeyen, R. Housley, and W. Polk. Internet x.509 public key infrastructure certificate and certificate revocation list (crl) profile. RFC 5280, RFC Editor, May 2008. <http://www.rfc-editor.org/rfc/rfc5280.txt>.
- [CSRT22] Jorge Chávez-Saab, Francisco Rodríguez-Henríquez, and Mehdi Tibouchi. Verifiable isogeny walks: Towards an isogeny-based postquantum VDF. In Riham AlTawy and Andreas Hülsing, editors, *SAC 2021: 28th Annual International Workshop on Selected Areas in Cryptography*, volume 13203 of *Lecture Notes in Computer Science*, pages 441–460, Virtual Event, September 29 – October 1, 2022. Springer, Cham, Switzerland.
- [CW05] John Cremona and Mark Watkins. Computing isogenies of elliptic curves, 2005. Preprint: <http://magma.maths.usyd.edu.au/users/watkins/papers/isogs.ps>.
- [DDGZ22] Luca De Feo, Samuel Dobson, Steven D. Galbraith, and Lukas Zobernig. SIDH proof of knowledge. In Shweta Agrawal and Dongdai Lin, editors, *Advances in Cryptology – ASIACRYPT 2022, Part II*, volume 13792 of *Lecture Notes in Computer Science*, pages 310–339, Taipei, Taiwan, December 5–9, 2022. Springer, Cham, Switzerland.
- [DFJP14] Luca De Feo, David Jao, and Jérôme Plût. Towards quantum-resistant cryptosystems from supersingular elliptic curve isogenies. *Journal of Mathematical Cryptology*, 8(3):209–247, 2014.
- [DFV24] Max Duparc, Tako Boris Fouotsa, and Serge Vaudenay. SILBE: An updatable public key encryption scheme from lollipop attacks. In Maria Eichlseder and Sébastien Gambs, editors, *SAC 2024: 31st Annual International Workshop on Selected Areas in Cryptography, Part I*, volume 15516 of *Lecture Notes in Computer Science*, pages 151–177, Montreal, Canada, August 28–30, 2024. Springer, Cham, Switzerland.
- [DKL⁺20] Luca De Feo, David Kohel, Antonin Leroux, Christophe Petit, and Benjamin Wesolowski. SQISign: Compact post-quantum signatures from quaternions and isogenies. In Shiho Moriai and Huaxiong Wang, editors, *Advances in Cryptology – ASIACRYPT 2020, Part I*, volume 12491 of *Lecture Notes in Computer Science*, pages 64–93, Daejeon, South Korea, December 7–11, 2020. Springer, Cham, Switzerland.
- [DMPS19] Luca De Feo, Simon Masson, Christophe Petit, and Antonio Sanso. Verifiable delay functions from supersingular isogenies and pairings. In Steven D. Galbraith and Shiho Moriai, editors, *Advances in Cryptology – ASIACRYPT 2019, Part I*, volume 11921 of *Lecture Notes in Computer Science*, pages 248–277, Kobe, Japan, December 8–12, 2019. Springer, Cham, Switzerland.
- [DOT21] Cyprien Delpech de Saint Guilhem, Emmanuela Orsini, and Titouan Tanguy. Limbo: Efficient zero-knowledge MPCitH-based arguments. In Giovanni Vigna and Elaine Shi, editors, *ACM CCS 2021: 28th Conference on Computer and Communications Security*, pages 3022–3036, Virtual Event, Republic of Korea, November 15–19, 2021. ACM Press.
- [Elk98] Noam D. Elkies. Elliptic and modular curves over finite fields and related computational issues. In D. A. Buell and J. T. Teitelbaum, editors, *Computational Perspectives on Number Theory*, volume 7 of *Studies in Advanced Mathematics*, pages 21–76. American Mathematical Society (AMS), 1998.
- [FMP23] Tako Boris Fouotsa, Tomoki Moriya, and Christophe Petit. M-SIDH and MD-SIDH: Countering SIDH attacks by masking information. In Carmit Hazay and Martijn Stam, editors, *Advances in Cryptology – EUROCRYPT 2023, Part V*, volume 14008 of *Lecture Notes in Computer Science*, pages 282–309, Lyon, France, April 23–27, 2023. Springer, Cham, Switzerland.

- [Fri11] Robert Fricke. *Die elliptischen Funktionen und ihre Anwendungen: Zweiter Teil: Die algebraischen Ausführungen*. Life Science and Basic Disciplines (German Language). Springer Berlin, Heidelberg, 1st edition, 2011. Originally published by B.G. Teubner-Verlag. (1922).
- [GHL⁺22] Tim Güneysu, Philip W. Hodges, Georg Land, Mike Ounsworth, Douglas Stebila, and Greg Zaverucha. Proof-of-possession for KEM certificates using verifiable generation. In Heng Yin, Angelos Stavrou, Cas Cremers, and Elaine Shi, editors, *ACM CCS 2022: 29th Conference on Computer and Communications Security*, pages 1337–1351, Los Angeles, CA, USA, November 7–11, 2022. ACM Press.
- [GKM⁺18] Jens Groth, Markulf Kohlweiss, Mary Maller, Sarah Meiklejohn, and Ian Miers. Updatable and universal common reference strings with applications to zk-SNARKs. In Hovav Shacham and Alexandra Boldyreva, editors, *Advances in Cryptology – CRYPTO 2018, Part III*, volume 10993 of *Lecture Notes in Computer Science*, pages 698–728, Santa Barbara, CA, USA, August 19–23, 2018. Springer, Cham, Switzerland.
- [GLS⁺23] Alexander Golovnev, Jonathan Lee, Srinath T. V. Setty, Justin Thaler, and Riad S. Wahby. Brakedown: Linear-time and field-agnostic SNARKs for R1CS. In Helena Handschuh and Anna Lysyanskaya, editors, *Advances in Cryptology – CRYPTO 2023, Part II*, volume 14082 of *Lecture Notes in Computer Science*, pages 193–226, Santa Barbara, CA, USA, August 20–24, 2023. Springer, Cham, Switzerland.
- [Gow90] Roderick Gow. Some properties and uses of the discriminant of a polynomial. In Ray Ryan, Ted Hurley, and Phil Rippon, editors, *Irish Mathematical Society Bulletin*, volume 24, pages 12–19. Irish Mathematical Society, March 1990.
- [GPS17] Steven D. Galbraith, Christophe Petit, and Javier Silva. Identification protocols and signature schemes based on supersingular isogeny problems. In Tsuyoshi Takagi and Thomas Peyrin, editors, *Advances in Cryptology – ASIACRYPT 2017, Part I*, volume 10624 of *Lecture Notes in Computer Science*, pages 3–33, Hong Kong, China, December 3–7, 2017. Springer, Cham, Switzerland.
- [Gro16] Jens Groth. On the size of pairing-based non-interactive arguments. In Marc Fischlin and Jean-Sébastien Coron, editors, *Advances in Cryptology – EUROCRYPT 2016, Part II*, volume 9666 of *Lecture Notes in Computer Science*, pages 305–326, Vienna, Austria, May 8–12, 2016. Springer Berlin Heidelberg, Germany.
- [JD11] David Jao and Luca De Feo. Towards quantum-resistant cryptosystems from supersingular elliptic curve isogenies. In Bo-Yin Yang, editor, *Post-Quantum Cryptography - 4th International Workshop, PQCrypto 2011*, pages 19–34, Tapei, Taiwan, November 29 – December 2 2011. Springer Berlin Heidelberg, Germany.
- [Kan89] Masanobu Kaneko. Supersingular j -invariants as singular moduli mod p . *Osaka Journal of Mathematics*, 26(4):849–855, December 1989.
- [Koh96] David Kohel. *Endomorphism rings of elliptic curves over finite fields*. PhD thesis, University of California, Berkeley, 1996. <https://www.i2m.univ-amu.fr/perso/david.kohel/pub/thesis.pdf>.
- [Lan87] Serge Lang. *Elliptic Functions*, volume 112 of *Graduate Texts in Mathematics*. Springer New York, NY, 1987.
- [LB20] Jonathan Love and Dan Boneh. Supersingular curves with small noninteger endomorphisms. In *ANTS XIV: Proceedings of the Fourteenth Algorithmic Number Theory Symposium*, volume 4 of *Open Book Series*, pages 7–22. Mathematical Sciences Publishers, 2020.
- [Ler97] Reynald Lercier. *Algorithmique des courbes elliptiques dans les corps finis*. PhD thesis, Ecole Polytechnique, June 1997. <https://univ-rennes.hal.science/tel-01101949>.

- [Lip22] Helger Lipmaa. A unified framework for non-universal SNARKs. In Goichiro Hanaoka, Junji Shikata, and Yohei Watanabe, editors, *PKC 2022: 25th International Conference on Theory and Practice of Public Key Cryptography, Part I*, volume 13177 of *Lecture Notes in Computer Science*, pages 553–583, Virtual Event, March 8–11, 2022. Springer, Cham, Switzerland.
- [LOX20] Songsong Li, Yi Ouyang, and Zheng Xu. Neighborhood of the supersingular elliptic curve isogeny graph at $j = 0$ and 1728. *Finite Fields and Their Applications*, 61:101600, 2020.
- [LP24] Shai Levin and Robi Pedersen. Faster proofs and VRFs from isogenies. *Cryptology ePrint Archive*, Paper 2024/1626, 2024.
- [Mes86] Jean-Francois Mestre. La méthode des graphes. exemples et applications. In *Proceedings of the international conference on class numbers and fundamental units of algebraic number fields (Katata)*, pages 217–242, 1986. English version (tr. Andrei Jorza): <https://wstein.org/papers/rank4/mestre-en.pdf>.
- [Mor95] François Morain. Calcul du nombre de points sur une courbe elliptique dans un corps fini: aspects algorithmiques. *Journal de Théorie des Nombres de Bordeaux*, 7(1):255–282, 1995.
- [Mor23] Tomoki Moriya. IS-CUBE: An isogeny-based compact KEM using a boxed SIDH diagram. *Cryptology ePrint Archive*, Paper 2023/1506, 2023.
- [Mü95] Volker Müller. *Ein Algorithmus zur Bestimmung der Punktzahl elliptischer Kurven über endlichen Körpern der Charakteristik größer drei*. PhD thesis, Universität des Saarlandes, 1995. <https://publikationen.sulb.uni-saarland.de/handle/20.500.11880/25777>.
- [NS22] Ngoc Khanh Nguyen and Gregor Seiler. Practical sublinear proofs for R1CS from lattices. In Yevgeniy Dodis and Thomas Shrimpton, editors, *Advances in Cryptology – CRYPTO 2022, Part II*, volume 13508 of *Lecture Notes in Computer Science*, pages 133–162, Santa Barbara, CA, USA, August 15–18, 2022. Springer, Cham, Switzerland.
- [Piz90] Arnold K. Pizer. Ramanujan graphs and Hecke operators. *Bull. Am. Math. Soc., New Ser.*, 23(1):127–137, 1990.
- [RS06] Alexander Rostovtsev and Anton Stolbunov. Public-key cryptosystem based on isogenies. *Cryptology ePrint Archive*, Paper 2006/145, 2006.
- [RY07] Thomas Ristenpart and Scott Yilek. The power of proofs-of-possession: Securing multi-party signatures against rogue-key attacks. In Moni Naor, editor, *Advances in Cryptology – EUROCRYPT 2007*, volume 4515 of *Lecture Notes in Computer Science*, pages 228–245, Barcelona, Spain, May 20–24, 2007. Springer Berlin Heidelberg, Germany.
- [Sil09] Joseph H. Silverman. *The arithmetic of elliptic curves*, volume 106 of *Grad. Texts Math.* Springer New York, NY, 2nd edition, 2009.
- [SSW20] Peter Schwabe, Douglas Stebila, and Thom Wiggers. Post-quantum TLS without handshake signatures. In Jay Ligatti, Xinming Ou, Jonathan Katz, and Giovanni Vigna, editors, *ACM CCS 2020: 27th Conference on Computer and Communications Security*, pages 1461–1480, Virtual Event, USA, November 9–13, 2020. ACM Press.
- [Sut13] Andrew V. Sutherland. Isogeny volcanoes. In Everett W. Howe and Kiran S. Kedlaya, editors, *ANTS X. Proceedings of the Tenth Algorithmic Number Theory Symposium*, volume 1 of *The Open Book Series*, pages 507–530. Mathematical Sciences Publishers, November 2013.
- [Tha22] Justin Thaler. Proofs, arguments, and zero-knowledge. *Foundations and Trends in Privacy and Security*, 4(2–4):117–660, 2022.

- [Tsu13] Kiminori Tsukazaki. *Explicit isogenies of elliptic curves*. PhD thesis, University of Warwick, July 2013. <http://webcat.warwick.ac.uk/record=b2688905-S1>.
- [vzGG13] Joachim von zur Gathen and Jürgen Gerhard. *Modern Computer Algebra*. Cambridge University Press, Cambridge, 3rd edition, 2013.
- [vzGL03] Joachim von zur Gathen and Thomas Lücking. Subresultants revisited. *Theoretical Computer Science*, 297(1):199–239, 2003. Latin American Theoretical Informatics.
- [Was08] Lawrence C. Washington. *Elliptic curves. Number theory and cryptography*. Chapman and Hall/CRC, Boca Raton, FL, 2nd edition, 2008.
- [XZS22] Tiancheng Xie, Yupeng Zhang, and Dawn Song. Orion: Zero knowledge proof with linear prover time. In Yevgeniy Dodis and Thomas Shrimpton, editors, *Advances in Cryptology – CRYPTO 2022, Part IV*, volume 13510 of *Lecture Notes in Computer Science*, pages 299–328, Santa Barbara, CA, USA, August 15–18, 2022. Springer, Cham, Switzerland.
- [XZZ⁺19] Tiancheng Xie, Jiaheng Zhang, Yupeng Zhang, Charalampos Papamanthou, and Dawn Song. Libra: Succinct zero-knowledge proofs with optimal prover computation. In Alexandra Boldyreva and Daniele Micciancio, editors, *Advances in Cryptology – CRYPTO 2019, Part III*, volume 11694 of *Lecture Notes in Computer Science*, pages 733–764, Santa Barbara, CA, USA, August 18–22, 2019. Springer, Cham, Switzerland.

A More on Elliptic Curves

A.1 Elliptic Curves and Isogenies

In this subsection we gather some well-known definitions and results on elliptic curves and isogenies. We fix a perfect field K .

An *elliptic curve* E over K is a non-singular projective algebraic curve of genus 1 with a distinguished ‘point at infinity’, usually denoted by O . The set of K -rational points $E(K)$ of such a curve forms a group with neutral element O .

Let E_0 and E_1 be elliptic curves over K , and $K \subseteq L \subseteq \bar{K}$ a field extension of K . An *isogeny* (defined over L , or *L -isogeny*), is a morphism of the curves $\phi: E_0 \rightarrow E_1$ over L (in particular it can be expressed by rational maps with coefficients in L) which induces also a surjective group homomorphism on the sets of $\bar{K}$ -rational points.

Any isogeny ϕ has a finite kernel, and the cardinality of this kernel equals the *degree* $\deg(\phi)$ of ϕ as a morphism if ϕ is separable [Sil09, Theorem III.4.10]. Further, the degree is multiplicative: If $\phi_1: E_0 \rightarrow E_1$ and $\phi_2: E_1 \rightarrow E_2$ are two isogenies, then $\deg(\phi_2 \circ \phi_1) = \deg(\phi_2) \cdot \deg(\phi_1)$.

Given an elliptic curve E_0 over K and any finite subgroup $G \subseteq E_0(\bar{K})$, there exist a unique (up to equivalence) elliptic curve E_1 and a separable isogeny $\phi_G: E_0 \rightarrow E_1$ with kernel equal to G [Sil09, Proposition III.4.12].

A classic example of an n^2 -isogeny is the *multiplication-by- n* endomorphism $[n]$ of E , which maps each $\bar{K}$ -rational point of an elliptic curve to its n^{th} scalar multiple. The kernel of the induced map on the $\bar{K}$ -rational points is called the *n -torsion* of E , denoted $E[n]$.

Notably, each n -isogeny $\phi: E_0 \rightarrow E_1$ admits a *dual isogeny* of degree n , which is the unique isogeny $\hat{\phi}: E_1 \rightarrow E_0$ such that $\phi \circ \hat{\phi} = \hat{\phi} \circ \phi = [n]$ [Sil09, Theorem III.6.1-2].

A.2 Canonical Modular Polynomials for $\kappa = 1$

Below we list the canonical modular polynomials for the primes $\ell \in \{2, 3, 5, 7, 13\}$, which are the primes that satisfy $\kappa = \frac{\ell-1}{\gcd(12, \ell-1)} = 1$.

$$\begin{aligned}\Phi_2^c(X, j) &= X^3 + 48X^2 + 768X + 4096 - X \cdot j, \\ \Phi_3^c(X, j) &= X^4 + 36X^3 + 270X^2 + 756X + 729 - X \cdot j, \\ \Phi_5^c(X, j) &= X^6 + 30X^5 + 315X^4 + 1300X^3 + 1575X^2 + 750X + 125 - X \cdot j, \\ \Phi_7^c(X, j) &= X^8 + 28X^7 + 322X^6 + 1904X^5 + 5915X^4 + 8624X^3 + 4018X^2 \\ &\quad + 748X + 49 - X \cdot j, \\ \Phi_{13}^c(X, j) &= X^{14} + 26X^{13} + 325X^{12} + 2548X^{11} + 13832X^{10} + 54340X^9 \\ &\quad + 157118X^8 + 333580X^7 + 509366X^6 + 534820X^5 + 354536X^4 \\ &\quad + 124852X^3 + 15145X^2 + 746X + 13 - X \cdot j.\end{aligned}$$

A.3 The Polynomial Factors of Lemma 7

In this subsection we list the polynomials g_{j^*} and $h_{j^*, \pm}$ described in Lemma 7; for $j^* = 0$ these polynomials are given in Table 6, and for $j^* = 1728$ in Table 7.

ℓ	g_0	$h_{0,+}$	$h_{0,-}$
2	1	$X + 16$	$X + 256$
3	$X + 27$	$X + 3$	$X + 243$
5	1	$X^2 + 10X + 5$	$X^2 + 250X + 3125$
7	$X^2 + 13X + 49$	$X^2 + 5X + 1$	$X^2 + 245X + 2401$
13	$X^2 + 5X + 13$	$X^4 + 7X^3 + 20X^2 + 19X + 1$	$X^4 + 247X^3 + 3380X^2 + 15379X + 28561$

Table 6: The polynomials g_0 and $h_{0,\pm}$.

ℓ	$g_{\ell,1728}$	$h_{\ell,1728,+}$	$h_{\ell,1728,-}$
2	$X + 64$	$X - 8$	$X - 512$
3	1	$X^2 + 18X - 27$	$X^2 - 486X - 19683$
5	$X^2 + 22X + 125$	$X^2 + 4X - 1$	$X^2 - 500X - 15625$
7	1	$X^4 + 14X^3 + 63X^2 + 70X - 7$	$X^4 - 490X^3 - 21609X^2 - 235298X - 823543$
13	$X^2 + 6X + 13$	$X^6 + 10X^5 + 46X^4 + 108X^3 + 122X^2 + 38X - 1$	$X^6 - 494X^5 - 20618X^4 - 237276X^3 - 1313806X^2 - 3712930X - 4826809$

Table 7: The polynomials g_{1728} and $h_{1728,\pm}$.

A.4 Rational Formulas for the Reconstruction Theorem

In this subsection we give, for the Reconstruction Theorem, the (ℓ -dependent) expressions of the root f as a K -rational function in the coefficients of $\phi_\ell(f)$. In these formulas we index f on the left hand side by the prime ℓ for emphasis.

For $\ell = 3$ we write $\phi_3(f) = x + C$ to obtain

$$f_3 = -\frac{(2A^2 + 7ACj_0 + 3C^3j_0)^2}{2^8 \cdot 3 \cdot A^3 \cdot j_0}.$$

For $\ell \in \{5, 7, 13\}$ we can use $\text{char}(K) \notin \{2, 3\}$ to compute $\text{disc}(\phi_\ell(f))$ by evaluating $\text{disc}(\phi_\ell(T))$ at $T = f$ according to Proposition 1(a). Thus computations in $\mathbb{Z}[T, T^{-1}]$ show

$$f_5 = \frac{(-48A)^3}{\text{disc}(\phi_5(f))^3 \cdot j_0} \text{ and } f_7 = \frac{(-48A)^3}{\text{disc}(\phi_7(f)) \cdot j_0}.$$

Lastly, for $\ell \in \{2, 13\}$ we see that the discriminant $\Delta = 2^{11} \cdot 3^5 \cdot f^{\ell-1} \cdot A \cdot B \neq 0$ of the target curve lies in $K(s_1, \dots, s_n)$ by Kohel's description of Vélú's formulas [Koh96, §2.4], and

$$f_2 = \frac{\Delta}{2^{11} \cdot 3^5 \cdot A \cdot B} \text{ and } f_{13} = \frac{2^{297} \cdot 3^{135} \cdot A^{27} \cdot B^{27}}{\text{disc}(\phi_{13}(f))^5 \cdot \Delta^2}.$$

B The Sylvester Matrix and the Resultant

In this section we discuss the theory of *resultants*, using [Bos18, §4.4] and [vzGG13, §6.3] as general references, together with novel applications that will be important for our proofs. Throughout, we fix R to be a commutative (as well as unital and associative) ring.

We briefly recall the adjugate matrix: Let $n \in \mathbb{N}$ and suppose that we have a square matrix $M = (m_{ij}) \in R^{n \times n}$. For any $i, j \in [n] = \{1, \dots, n\}$ we let M_{ij} denote the matrix obtained from M by removing the i^{th} row and the j^{th} column. Then the *adjugate matrix* $\text{adj}(M)$ of M is defined as the square matrix $\text{adj}(M) = (a_{ij})_{i,j=1}^n \in R^{n \times n}$ with entries

$$a_{ij} = (-1)^{i+j} \det(M_{ji}).$$

The following result gives the well known *Laplace expansion* formulas:

Lemma 13. We have

$$\text{adj}(M) \cdot M = \det(M) I_n = M \cdot \text{adj}(M)$$

where I_n denotes the n^{th} identity matrix. Equivalently, for any $i \in [n]$ we can compute $\det(M)$ via *Laplace expansion along the i^{th} row* given by

$$\det(M) = \sum_{j=1}^n (-1)^{i+j} \cdot m_{ij} \cdot \det(M_{ij}),$$

or via Laplace expansion along the i^{th} column given by

$$\det(M) = \sum_{j=1}^n (-1)^{j+i} \cdot m_{ji} \cdot \det(M_{ji}).$$

For $n \in \mathbb{N}_0$ we now consider the free R -module

$$\mathcal{P}_n := \{s \in R[X] : \deg(s) < n\}$$

of rank n , equipped with the *descending monomial basis* $(X^{n-1}, X^{n-2}, \dots, X, 1)$.

Definition 2. Let $g, h \in R[X]$ be non-zero polynomials of respective degrees $d, e \in \mathbb{N}_0$ and define the R -linear map

$$\sigma : \mathcal{P}_e \times \mathcal{P}_d \rightarrow \mathcal{P}_{d+e}, (s, t) \mapsto s \cdot g + t \cdot h.$$

We call the representation matrix of σ with respect to the descending monomial bases of each $\mathcal{P}_n$ the *Sylvester matrix* $\text{Syl}(g, h)$ of g and h .

Remark 7. In the literature the transpose of the representation matrix of σ is sometimes referred to as the Sylvester matrix instead – see, for example, [BPR06, Notation 4.12].

Example 3. For the polynomials $g = a_0X^4 + a_1X^3 + a_2X^2 + a_3X + a_4$ and $h = b_0X^2 + b_1X + b_2$ with $a_0, b_0 \neq 0$ we have

$$\text{Syl}(g, h) = \begin{pmatrix} a_0 & & & & b_0 & & & \\ a_1 & a_0 & b_1 & b_0 & & & & \\ a_2 & a_1 & b_2 & b_1 & b_0 & & & \\ a_3 & a_2 & & b_2 & b_1 & b_0 & & \\ a_4 & a_3 & & & b_2 & b_1 & b_0 & \\ & a_4 & & & & b_2 & b_1 & b_0 \end{pmatrix},$$

where empty entries are understood to be zero.

Working over a field $R = K$, the kernel of σ can be given explicitly, and the knowledge of its codimension, i.e. of the rank of $\text{Syl}(g, h)$, will be crucial later:

Lemma 14. Let K be a field and let $g, h \in K[X]$ be non-zero polynomials. Then the kernel of σ is given by

$$\ker(\sigma) = \left\{ \left(\frac{h}{\gcd(g, h)} \cdot r, -\frac{g}{\gcd(g, h)} \cdot r \right) \mid r \in K[X] \text{ with } \deg(r) < \deg \gcd(g, h) \right\}.$$

In particular, the rank of the Sylvester matrix of g and h satisfies

$$\text{rk Syl}(g, h) = \deg(g) + \deg(h) - \deg \gcd(g, h).$$

Proof. Let us write $\gamma := \gcd(g, h)$. For any $r \in K[X]$ with $\deg(r) < \deg(\gamma)$ we have

$$\deg(h/\gamma \cdot r) = \deg(h) - \deg(\gamma) + \deg(r) < \deg(h)$$

and similarly $\deg(-g/\gamma \cdot r) < \deg(g)$. Furthermore

$$\sigma(h/\gamma \cdot r, -g/\gamma \cdot r) = r \cdot (h/\gamma \cdot g - g/\gamma \cdot h) = r \cdot \gamma^{-1} \cdot (hg - gh) = 0,$$

which shows that $\ker(\sigma)$ contains all elements of the form $(h/\gamma \cdot r, -g/\gamma \cdot r)$.

Conversely consider an element $(s, t) \in \ker(\sigma)$, so that $s \cdot g = (-t) \cdot h$. Dividing by γ then yields $s \cdot g/\gamma = (-t) \cdot h/\gamma$, and since g/γ and h/γ are coprime we find $r_s, r_t \in K[X]$ with $s = h/\gamma \cdot r_s$ and $t = g/\gamma \cdot r_t$. Plugging these values into σ now gives

$$0 = \sigma(s, t) = h/\gamma \cdot r_s \cdot g + g/\gamma \cdot r_t \cdot h = (r_s + r_t) \cdot gh \cdot \gamma^{-1},$$

so we have $r_t = -r_s$, i.e. $t = -g/\gamma \cdot r_s$. Due to

$$\deg(r_s) = \deg(s) - \deg(h) + \deg(\gamma) < \deg(\gamma)$$

we thus get the other claimed inclusion by setting $r := r_s$. Finally, the rank formula for $\text{Syl}(g, h)$ now follows directly from the rank-nullity theorem since $\ker(\sigma)$ has K -dimension $\deg \gcd(g, h)$. $\square$

The determinant of the Sylvester matrix also plays an important role:

Definition 3. Let $g, h \in R[X]$ be non-zero polynomials. The *resultant* $\text{res}(g, h)$ of g and h is defined as

$$\text{res}(g, h) := \det \text{Syl}(g, h).$$

For emphasis we sometimes indicate the variable X with respect to which the resultant is taken in the index of res – for example, we would write $\text{res}_X(g, h)$ above.

By Lemma 14 the rank of the Sylvester matrix is maximal for coprime polynomials, and this can alternatively be expressed via the resultant:

Corollary 15. Let K be a field and let $g, h \in K[X]$ be non-zero polynomials. Then the following are equivalent:

- (i) g and h are coprime, i.e. $\gcd(g, h) = 1$.
- (ii) $\text{rk Syl}(g, h) = \deg(g) + \deg(h)$.
- (iii) $\text{res}(g, h) \neq 0$.

Proof. By definition $\text{Syl}(g, h)$ is a square matrix with $\deg(g) + \deg(h)$ rows, so the equivalence of (ii) and (iii) follows from linear algebra. Moreover, Lemma 14 shows that (i) and (ii) are equivalent. $\square$

Next we note an immediate but important compatibility result that we will use frequently in the sequel:

Lemma 16. Let $g, h \in R[X]$ be non-zero polynomials, and let $\varphi: R \rightarrow S$ be a ring homomorphism, extended to a ring homomorphism $\varphi: R[X] \rightarrow S[X]$ via coefficient-wise application. If φ preserves the degrees of g and h , then we have

$$\varphi(\text{Syl}(g, h)) = \text{Syl}(\varphi(g), \varphi(h)),$$

where on the left hand side φ is applied entry-wise. In particular, in this situation we have

$$\varphi(\text{res}(g, h)) = \text{res}(\varphi(g), \varphi(h)).$$

Proof. This follows from the definitions and the fact that the determinant is compatible with ring homomorphisms as it is defined as a multivariate polynomial in the matrix entries. $\square$

This compatibility result also extends Corollary 15 to integral domains:

Corollary 17. Let R be an integral domain and let $g, h \in R[X]$ be non-zero polynomials. Then g and h share a common divisor of positive degree if and only if $\text{res}(g, h) = 0$.

Proof. We consider the embedding $\varphi: R \rightarrow K$ of R into its field of fractions K , and its extension to $R[X] \rightarrow K[X]$ via coefficient-wise application. As this clearly preserves the degrees of g and h , we see with Lemma 16 that $\text{res}(g, h)$ is non-zero if and only if $\text{res}(\varphi(g), \varphi(h))$ is. Furthermore, g and h share no common divisor of positive degree if and only if $\varphi(g)$ and $\varphi(h)$ are coprime, as both statements are equivalent to the fact that $\varphi(g)$ and $\varphi(h)$ do not have a common root in an algebraic closure of K . Therefore the claim follows from Corollary 15. $\square$

For our proofs we want to extend Corollary 17 to relate the number of common roots (i.e. the maximal degree of a common divisor) to the resultant. While prior work in this direction using subresultants dates far back (see [vzGL03, §1.1] for a historical overview), we want to keep our focus on the resultant. To do so, we will instead consider the situation that $R = A[Y]$ is itself a polynomial ring, and we will be interested in deriving the resultant $\text{res}_X(g, h) \in R$ with respect to Y . To connect for this analysis the derivatives to k -minors of $\text{Syl}(g, h)$, i.e. to determinants of $(k \times k)$ -submatrices of $\text{Syl}(g, h)$, we will use *Jacobi's formula* as our main tool:

Lemma 18 (Jacobi). Let $R = A[Y]$ be a polynomial ring over a commutative ring A , let $n \in \mathbb{N}$ and let $M = (m_{ij}) \in R^{n \times n}$ be a square matrix. Furthermore let $\frac{\partial}{\partial Y} M$ denote the matrix obtained from M via entry-wise derivation. Then

$$\frac{\partial}{\partial Y} \det(M) = \text{tr} \left(\text{adj}(M) \cdot \frac{\partial}{\partial Y} M \right).$$

In particular, for any $k \in \{0, \dots, n\}$ we have

$$\frac{\partial^k}{\partial Y^k} \det(M) \in R \cdot \{(n-k)\text{-minors of } M\},$$

i.e. the k^{th} derivative of $\det(M)$ with respect to Y is an R -linear combination of $(n-k)$ -minors of M .

Proof. For ease of notation we index submatrices of M by indices in $[n] \times [n]$ where we disallow indices of removed rows as the first index respectively of removed columns as the second index. We proceed by induction on n : For $n = 1$ the matrix $\text{adj}(M)$ has the single entry 1, so the formula clearly holds. Assuming $n \geq 2$, we apply the product rule to the Laplace expansion of $\det(M)$ along the last column to obtain

$$\frac{\partial}{\partial Y} \det(M) = \sum_{i=1}^n (-1)^{i+n} \det(M_{in}) \frac{\partial}{\partial Y} m_{in} + (-1)^{i+n} m_{in} \frac{\partial}{\partial Y} \det(M_{in}). \quad (11)$$

By the induction hypothesis we furthermore have

$$\begin{aligned} \frac{\partial}{\partial Y} \det(M_{in}) &= \text{tr} \left(\text{adj}(M_{in}) \cdot \frac{\partial}{\partial Y} M_{in} \right) \\ &= \sum_{j=1}^{n-1} \sum_{l=1}^{i-1} (-1)^{j+l} \det((M_{in})_{lj}) \frac{\partial}{\partial Y} m_{lj} + \sum_{j=1}^{n-1} \sum_{l=i+1}^n (-1)^{j+l-1} \det((M_{in})_{lj}) \frac{\partial}{\partial Y} m_{lj}. \end{aligned}$$

Therefore swapping the summation order of l and i yields

$$\begin{aligned} &\sum_{i=1}^n (-1)^{i+n} m_{in} \frac{\partial}{\partial Y} \det(M_{in}) \\ &= \sum_{j=1}^{n-1} \sum_{l=1}^n \sum_{i=l+1}^n (-1)^{l+j} (-1)^{(i-1)+(n-1)} m_{in} \det((M_{lj})_{in}) \frac{\partial}{\partial Y} m_{lj} \\ &\quad + \sum_{j=1}^{n-1} \sum_{l=1}^n \sum_{i=1}^{l-1} (-1)^{l+j} (-1)^{i+(n-1)} m_{in} \det((M_{lj})_{in}) \frac{\partial}{\partial Y} m_{lj} \\ &= \sum_{j=1}^{n-1} \sum_{l=1}^n (-1)^{l+j} \det(M_{lj}) \frac{\partial}{\partial Y} m_{lj} \end{aligned}$$

where we used Laplace expansion of $\det(M_{lj})$ along the last column to get rid of the sum over i . Now we see that the first summands in Equation (11) give precisely the n^{th} outer sum above, so in total we obtain:

$$\frac{\partial}{\partial Y} \det(M) = \sum_{j=1}^n \sum_{i=1}^n (-1)^{i+j} \det(M_{ij}) \frac{\partial}{\partial Y} m_{ij} = \text{tr} \left(\text{adj}(M) \cdot \frac{\partial}{\partial Y} M \right) \quad (12)$$

Finally we argue why the second claim follows from this formula by induction on k . For $k = 0$ the claim is immediate as $\det(M)$ is the unique $(n-0)$ -minor of M . Now expressing $\frac{\partial^{k-1}}{\partial Y^{k-1}} \det(M)$ as an R -linear combination of $(n-k+1)$ -minors of M via the induction hypothesis, we see by Jacobi's formula (12) (applied to each $(n-k+1)$ -minor of M) and the product rule that $\frac{\partial^k}{\partial Y^k} \det(M)$ is an R -linear combination of $(n-k+1)$ -minors of M and their $(n-k+1-1)$ -minors; the latter are $(n-k)$ -minors of M , and the former are R -linear combinations of $(n-k)$ -minors of M due to Laplace expansion, hence yielding the claim. $\square$

The following consequence is tailored to our needs:

Corollary 19. Let $R = K[Y]$ be a polynomial ring over a field K and fix an element $y_0 \in K$. Additionally let $g, h \in R[X]$ be non-zero polynomials and extend the K -linear evaluation homomorphism $\varphi: R \rightarrow K$ given by $Y \mapsto y_0$ to a ring homomorphism $\varphi: R[X] \rightarrow K[X]$ via coefficient-wise application. Further suppose that φ preserves the X -degrees of g and h , and write

$$m := \deg \gcd(\varphi(g), \varphi(h)).$$

Then

$$\left. \frac{\partial^k}{\partial Y^k} \right|_{Y=y_0} \text{res}_X(g, h) = 0 \text{ for } k \in \{0, \dots, m-1\}.$$

Proof. Let $k \in \{0, \dots, m-1\}$ and set $n := \deg(g) + \deg(h)$. By Lemma 18 the k^{th} Y -derivative of $\text{res}_X(g, h)$ is an R -linear combination of $(n-k)$ -minors of $\text{Syl}(g, h)$. Moreover, by Lemma 16 we have $\text{Syl}(\varphi(g), \varphi(h)) = \varphi(\text{Syl}(g, h))$, so the images of the $(n-k)$ -minors of $\text{Syl}(g, h)$ under φ are $(n-k)$ -minors of $\text{Syl}(\varphi(g), \varphi(h))$.

Hence

$$\varphi \left(\left. \frac{\partial^k}{\partial Y^k} \text{res}(g, h) \right|_{Y=y_0} \right) = \left. \frac{\partial^k}{\partial Y^k} \right|_{Y=y_0} \text{res}_X(g, h)$$

is a K -linear combination of $(n-k)$ -minors of $\text{Syl}(\varphi(g), \varphi(h))$. Finally, by Lemma 14 we know that

$$\text{rk } \text{Syl}(\varphi(g), \varphi(h)) = \deg(\varphi(g)) + \deg(\varphi(h)) - m = n - m < n - k,$$

so all $(n-k)$ -minors of $\text{Syl}(\varphi(g), \varphi(h))$ are zero by linear algebra and the claim follows. $\square$

To turn our attention to the second application of resultants in this paper, we relate the resultant $\text{res}(g, h)$ back to g and h :

Lemma 20. Let R be an integral domain and let $g, h \in R[X]$ be non-zero polynomials of respective degrees $d, e \in \mathbb{N}_0$ such that $d + e \geq 1$. Then there are polynomials $(s, t) \in \mathcal{P}_e \times \mathcal{P}_d$ such that

$$\text{res}(g, h) = sg + th.$$

Proof. Recalling the definition of $\text{res}(g, h)$ via the linear map $\sigma: \mathcal{P}_e \times \mathcal{P}_d \rightarrow \mathcal{P}_{d+e}$ and translating the existence of the adjugate matrix into linear maps, we obtain an R -linear map $\phi: \mathcal{P}_{d+e} \rightarrow \mathcal{P}_e \times \mathcal{P}_d$ such that

$$\sigma \circ \phi(r) = \det(\text{Syl}(g, h)) \cdot r = \text{res}(g, h) \cdot r$$

for all $r \in \mathcal{P}_{d+e}$. Applying this composition to $r = 1$, which is possible since $d + e \geq 1$, hence shows that $\phi(1) = (s, t)$ satisfies the required linear combination. $\square$

With this we can finish arguing the missing part of the proof of Theorem 9:

Proposition 21. Let $\ell \leq 13$ be a prime and let p_ℓ be given according to

$$(p_2, p_3, p_5, p_7, p_{11}, p_{13}) = (13, 53, 379, 1217, 5101, 8387).$$

Additionally let K be a field of characteristic $\text{char}(K) \notin [p_\ell]$ and let $j_0 \in K^\times$ be a non-zero j -invariant. Then $\Phi_\ell(j_0, Y)$ does not have a triple root in $\overline{K}$.

Proof. This proof is highly computational – the observational claims used along the way can be verified via the SageMath script `maximal_primes.sage`. Using multiple resultant computations, we derive a contradiction by considering the derivatives of the classical modular polynomial with respect to the second variable and applying the ring homomorphism

$$\rho: \mathbb{Z}[X][Y] \rightarrow K[Y], \quad X \mapsto j_0, \quad Y \mapsto Y.$$

For $a \in \{0, 1, 2\}$ let us write

$$\Psi_a(X, Y) := \frac{\partial^a}{\partial Y^a} \Phi_\ell(X, Y) \in \mathbb{Z}[X][Y]$$

and note that ρ preserves the Y -degree of each Ψ_a since $p_\ell > \ell$.

Suppose now that $\Phi_\ell(j_0, Y) \in K[Y]$ has a triple root; then the three polynomials $\rho(\Psi_0) = \Phi_\ell(j_0, Y)$, $\rho(\Psi_1)$ and $\rho(\Psi_2)$ all share a common root. Thus Lemma 16 and Corollary 15 show that the resultants

$$g_1 := \text{res}_Y(\Psi_0, \Psi_1), \quad g_2 := \text{res}_Y(\Psi_0, \Psi_2), \quad g_3 := \text{res}_Y(\Psi_1, \Psi_2) \in \mathbb{Z}[X]$$

satisfy $\rho(g_i) = 0$. As powers of j_0 and primes not larger than p_ℓ are invertible in K , we can divide out powers of X and such prime factors from each g_i to obtain new polynomials – which we will, to simplify notation, again denote by g_i – that still get sent to 0 by ρ .

Due to these modifications the three polynomials (g_1, g_2, g_3) turn out to pairwise have no common factor of positive degree, i.e. the number

$$\gamma_\ell := \gcd(\text{res}_X(g_1, g_2), \text{res}_X(g_1, g_3), \text{res}_X(g_2, g_3)) \in \mathbb{Z}$$

is non-zero by Corollary 17, and we can see that p_ℓ is its largest prime factor.

However, by Lemma 20 (noting that each g_i is non-constant) we find for any $i, j \in \{1, 2, 3\}, i < j$, polynomials $s_{ij}, t_{ij} \in \mathbb{Z}[X]$ with $\text{res}_X(g_i, g_j) = s_{ij}g_i + t_{ij}g_j$ and thus

$$\rho(\text{res}_X(g_i, g_j)) = \rho(s_{ij})\rho(g_i) + \rho(t_{ij})\rho(g_j) = 0.$$

Hence each $\text{res}_X(g_i, g_j)$ is zero in K , so $\text{char}(K) = p > 0$ has to be a prime factor of γ_ℓ . As p_ℓ is the maximal prime factor of γ_ℓ , we obtain a contradiction to our assumption on $\text{char}(K)$. $\square$

An important special case of the resultant is the *discriminant*, which we will define now: Let R be an integral domain and let $g \in R[X]$ such that $\frac{\partial}{\partial X}g$ is non-zero. Then all entries in the first row of $\text{Syl}(g, \frac{\partial}{\partial X}g)$ are divisible by the leading coefficient a_0 of g ; therefore the resultant $\text{res}(g, \frac{\partial}{\partial X}g)$ is also divisible by a_0 due to Laplace expansion along this first row, and one defines

$$\text{disc}(g) := (-1)^{\binom{\deg(g)}{2}} \cdot a_0^{-1} \cdot \text{res}\left(g, \frac{\partial}{\partial X}g\right) \in R$$

to be the *discriminant* of g . We directly obtain the following from Lemma 16:

Corollary 22. Let R be an integral domain and $g \in R[X]$ a polynomial such that $\frac{\partial}{\partial X}g$ is non-zero. Furthermore let $\varphi: R \rightarrow S$ be a ring homomorphism of integral domains, extended to a ring homomorphism $\varphi: R[X] \rightarrow S[X]$ via coefficient-wise application. If we have $\deg(g) = \deg(\varphi(g))$ and $\deg(\frac{\partial}{\partial X}g) = \deg(\frac{\partial}{\partial X}\varphi(g))$, then

$$\varphi(\text{disc}(g)) = \text{disc}(\varphi(g)).$$